AERead: Evaluating Economic Decisions in Interactive Environments

Chenyu Li · Zeyu Sun

contact@aeread.org

Research working paper · version 15.39 · 21 October 2026 · Preliminary results; feedback and contributions welcome

Abstract

AI agents are increasingly asked to make economic decisions for a principal, such as whom to trade with, what to verify and when to commit. In these decisions a completed transaction can destroy value and a refusal can be correct, so task-completion or agreement scores cannot tell whether an agent served its principal. We present AERead, an environment that scores these decisions by the value an agent obtains. Each case defines a distribution of economic worlds, seats with private information, counterparts with their own interests, and a reference policy whose access to information is stated, so that a shortfall separates poor decisions from information the agent never had. One runtime executes every case and recomputes each score from the sealed record without calling a model, and quality control operates on populations of worlds rather than on single tasks. We build cases in housing, procurement, data-center contracting and refunds on this runtime. The counterpart is part of the measurement: the earnings gap between two agents is 42% of the value available in a deal when they negotiate against each other, and 7% when each faces the same scripted counterpart. We break shortfalls down by stratum and by action class; in housing, nearly all of the gap to the reference falls in worlds where the tenant's favourite listing is a lemon. Of 25 defects selected from our quality-control record, 14 were invisible in any one world. Housing and Procurement now pass four of the five quality-control gates and no case passes all five, so our results compare named agents on stated worlds rather than rank models. AERead's code is open source: a new case supplies its economic rules, worlds and reference, and inherits the runtime, replay and the quality-control standard.

1. Introduction

An agent buying on behalf of a firm must decide more than whether it can place an order. It must choose whom to trust, whether another test is worth its cost, and when to commit before an offer expires. A tenant faces the same problem when inspecting a home; a negotiator faces it when exchanging price for contractual risk. In these settings, completing a transaction can destroy value, while refusing it can be the right decision. We use *economic-agent readiness* to mean an agent's ability to obtain value for a principal within a stated mandate, given limited information and other parties' interests.

Evaluating this ability requires a reference for what the agent could have achieved. A good realized outcome can follow a poor decision, and a shortfall from a fully informed optimum can reflect information the agent never had. The counterpart also matters: a strategy that succeeds against a cooperative seller may fail against a different policy. Existing work provides interactive task evaluation and economic decision benchmarks (Merrill et al. 2026; Fish et al. 2025; Zhang et al. 2026). The problem we address is how to make information, counterpart behavior and reference values explicit across different economic cases, then test whether the resulting scores measure the intended decisions.

AERead provides a framework for constructing and evaluating such cases. Each case specifies the parties, their observations and legal actions, the world distribution, and checks on behavior and outcomes. A shared runtime executes play and records evidence. Case-specific references give scores an economic interpretation, and results keep their native units.

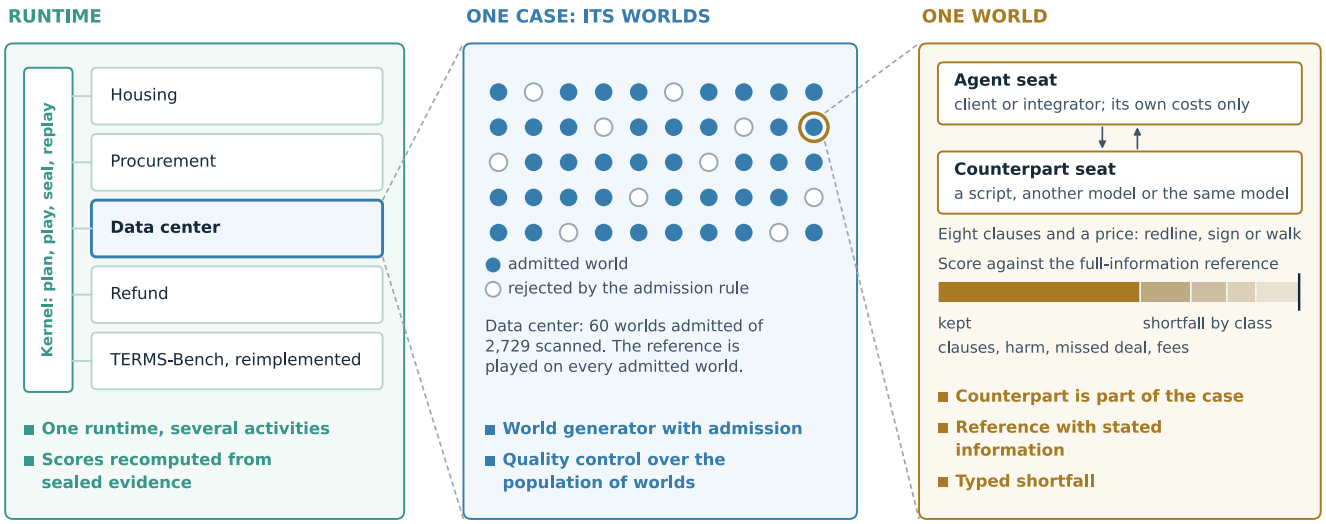


Figure 1.1. The three levels of an AERead evaluation, with the Data-center agreement as the example. The runtime hosts several case families (left). A case draws a population of worlds and admits those where the decision matters; quality control runs over that population (centre). In one world, the agent holds a seat, a counterpart holds another, and the score is compared with the case reference and split into shortfall classes (right). Dashed lines mean "contains", not a sequence. Each panel lists the properties of Section 1 that belong to its level. The dot pattern and the score bar are schematic.

AERead combines seven properties. Prior benchmarks have some of them, and TERMS-Bench has five in full. AERead has six in full and has the seventh, an inclusion criterion that screens out shortcuts, in three of its cases (Table 2.1). The first three properties concern a single world, the next two a case's population of worlds, and the last two the runtime (Figure 1.1). We built each of them and applied it to the cases reported here.

1. **Value against a reference with stated information.** Scores are economic values in each case's own units, measured against a feasible policy, the best policy given the deciding seat's information, or a labelled full-information bound. We play each reference through the environment on every included world (Section 6).
2. **The counterpart is part of the case.** Other parties are seats with their own information and interests, played by scripted policies, another model or the same model, and each claim names its counterpart. In the Data-center case, the measured gap between two agents depends strongly on which counterpart they face (Section 5).
3. **Typed shortfall.** The gap to the reference is split into observed action classes, such as tests not taken, worse terms, harmful agreements and negotiation fees, so a result states where value was lost as well as how much (Section 5).
4. **World generators with an inclusion criterion.** A case is a generator of worlds, and its evaluation sets regenerate from their seeds. An inclusion criterion, computed before any agent plays, keeps only worlds in which simple shortcuts lose a stated margin. The Data-center comparison, for example, used 80 worlds included from 5,436 scanned (Section 5).
5. **Quality control over populations of worlds.** Cases are qualified through quality-control gates (Section 6). We score simple-rule ceilings beside the agents and audit the task prompts. More than half of the defects we selected from our record were invisible in any single world.
6. **Scores recomputed from sealed evidence.** The kernel hashes a plan before execution, seals the events, and replays each receipt to recompute its scores without calling a model; no model judges a score (Section 4).

7. **One runtime for several activities.** Families supply only the economic rules. Four cases and a reimplementation of three TERMS-Bench counterpart families run on the same kernel (Section 5.3).

A researcher studying another economic decision therefore writes only its rules, worlds and reference. Execution, sealing, replay, counterpart seats and the quality-control gates come with the runtime, so a new case starts with the same evidence and checks as the existing ones. Sections 3 and 4 describe the case design and the runtime, Section 5 the cases, and Section 6 the quality control. Section 7 states what the results do and do not establish.

2. Related work

Table 2.1 compares the closest benchmarks on the seven properties of Section 1; the paragraphs below discuss them by theme.

Table 2.1. The seven properties in related benchmarks, read from each paper's description of its own method: ● present, ◐ partly present, – absent. Each cell was read against a written definition, and the lower mark was chosen where a reading was borderline; page references for every cell are kept with the source. AERead's row describes the cases reported here; Housing v15 passes four of the five qualification gates and none passes all five (Section 6). TERMS-Bench has five of the seven properties in full. AERead adds several activities on one runtime and, in three of its cases, an admission rule that screens shortcuts.

Benchmark	1. Value against a reference with stated information	2. Counterpart is part of the case	3. Typed shortfall	4. World generator with admission	5. Quality control over populations of worlds	6. Scores recomputed from sealed evidence	7. Several economic activities on one runtime
Terminal-Bench	–	–	◐	◐	●	◐	◐
τ²-bench	–	◐	◐	◐	◐	◐	◐
GDPval	–	–	◐	–	◐	–	◐
Agents' Last Exam	–	–	◐	◐	◐	◐	◐
EconEvals	●	–	◐	◐	◐	◐	●
NegotiationArena	◐	●	◐	–	–	◐	◐
AgenticPay	◐	●	◐	–	–	◐	◐
TERMS-Bench	●	●	●	◐	●	●	◐
AERead (this paper)	●	●	●	◐	●	●	●

Economically valuable work.

GDPval selects professional tasks by occupation and evaluates deliverables through expert comparison (Patwardhan et al. 2025). Agents' Last Exam (ALE) evaluates expert-authored professional workflows with task-specific verification (Sun et al. 2026). AERead evaluates decisions under a principal's mandate, with responsive counterparts and payoff references that specify the deciding party's information.

Interactive agent evaluation.

Terminal-Bench verifies final environment state and supplies human-written solutions (Merrill et al. 2026). AppWorld combines an application engine with state-based evaluation, including collateral-change checks (Trivedi et al. 2024). τ²-bench lets agent and user act on shared state (Barres et al. 2025).

Economic decisions and negotiation.

EconEvals covers procurement, scheduling, pricing and economic choice (Fish et al. 2025). NegotiationArena studies bargaining tactics (Bianchi et al. 2024); SidConArena combines negotiation, production and auctions (Feng, Chen, and He 2026). AgenticPay evaluates buyer–seller bargaining under private reservation values, scoring agreement, surplus allocation and speed (Liu, Gu, and Song 2026). TERMS-Bench specifies counterpart types, policies and payoffs to diagnose negotiation through oracle-reference gaps (Zhang et al. 2026); our case specifications follow its decomposition into environment, protocol and counterpart policy (Appendix A), and Section 5.3 reimplements three of its counterpart families on AERead's runtime. AERead retains case-specific payoff units, distinguishes full-information bounds from information-constrained policy references, and applies quality control to world distributions and measurement paths.

Shared environments and evaluation quality.

OpenSpiel and PettingZoo provide common interfaces for games and multi-agent environments (Lanctot et al. 2019; Terry et al. 2021). AERead similarly separates shared execution from case semantics (Appendix H). Terminal-Bench combines executable reference solutions, automated checks, human review and adversarial exploit testing (Merrill et al. 2026). The Agentic Benchmark Checklist identifies task and reward defects that distort performance (Zhu et al. 2025). Our QC analysis tests whether prompts, controls and counterpart assignments support valid comparisons across distributions of economic worlds.

---

3. Task design and construction

Three of AERead's properties are fixed when a case is designed: the reference and the information it may use, the counterpart seats, and the world generator with its inclusion criterion.

3.1 Cases, worlds and seats

A *case* defines an economic decision problem; a *world* is one instance of that design, with particular valuations, private types, costs and deadlines. A *seat* is a party's position in the interaction, with its own objective, information and permitted actions. A policy occupies a seat and chooses actions from its observations and available history. The family implementation supplies the environment, observation rules, legal actions, transitions and verifier.

For example, Housing places a tenant among four listings with hidden quality. The tenant can inspect, make offers and decide whether to sign; landlord replies and outside demand affect the opportunities that remain. Inspection changes both information and cost. The verifier therefore needs to distinguish a legal action, a favorable realized outcome and a good decision given the tenant's information.

In a controlled comparison, the evaluated policy occupies one seat and counterparts follow fixed policies. Cross-play and self-play change those assignments and answer different questions. Each comparison names the seat, counterpart configuration and world distribution to which it applies. Appendices A–D give the case specifications, counterpart policies, references and measures.

3.2 Selecting economic decisions

We organize coverage around activities in which a party's choices affect the value of a transaction (Li 2026). A candidate case needs a consequential decision, a counterpart with its own interests and information, and an outcome that can be evaluated inside the world. The intended coverage spans business purchases, labour contracts, capital and corporate transactions, household purchases, and public purchases. Contractual terms, risk allocation and disputes enter as decisions within these transactions.

The four examples cover business and household purchases. They were selected to exercise different references and counterpart configurations; they are not a representative sample of economic activity. Economic stakes motivate which decisions to study, but the reported scores are case-specific quantities, without a cross-family aggregate or an estimate of economy-wide gains.

3.3 Constructing and qualifying a case

A contribution begins with a decision and an interpretation of success. The case author specifies the principal's objective and constraints, what each seat can observe, which actions commit resources, and how counterpart policies respond. The task prompt must state the rules needed to make the decision while preserving the intended private information. These choices define the mechanism that a score is meant to test.

The author then provides a world generator or a fixed evaluation set, an inclusion criterion, and a reference whose access to information is explicit. The inclusion criterion screens each candidate world before any agent plays and before hidden types are drawn. It is computed exactly from the public world and the stated prior. It keeps a world only if simple shortcuts, such as accepting the default terms or ignoring private information, fall short of the reference by a stated margin. Whether the reference can be reached is a separate check. Executable controls play the reference through the actual environment and verifier on every included world. Simple policies are played beside it, to reveal shortcuts or ceiling effects that the screen missed. The evaluation set must also be valid, free of duplicates, and split into disjoint development and holdout ranges. A full-information bound is labelled separately from a policy the evaluated seat could execute. The case states its score units, the direction of each score, and its treatment of invalid actions and incomplete episodes.

Before a confirmatory comparison, the campaign freezes the worlds, seat assignments, prompts, agent settings, budgets and analysis, and pre-registers them. Its evidence package should connect each reported measure to the retained actions and outcomes, the scoring implementation, and a command that reproduces it. Section 4 describes the shared execution path. The current drivers and headline analyses extend beyond what a receipt covers in places, which Appendix H records.

This is the contribution and qualification standard. The four examples meet it to different degrees: their implementation support and completed checks vary (Section 6). Cases can be useful for development before qualification, provided their reports identify missing checks, preserve failed attempts, and distinguish analyses preregistered before execution from later corrections. The current examples demonstrate the workflow and its failure modes; they do not establish successful adoption by external contributors.

---

4. Shared runtime and evaluation harness

The runtime provides two of AERead's properties: one kernel executes every family, and every score is recomputed by code from sealed evidence. The shared runtime, or *kernel*, executes the case designs in Section 3. Families supply the economic rules; the kernel schedules play and records the evidence used for evaluation (Figure 4.1).

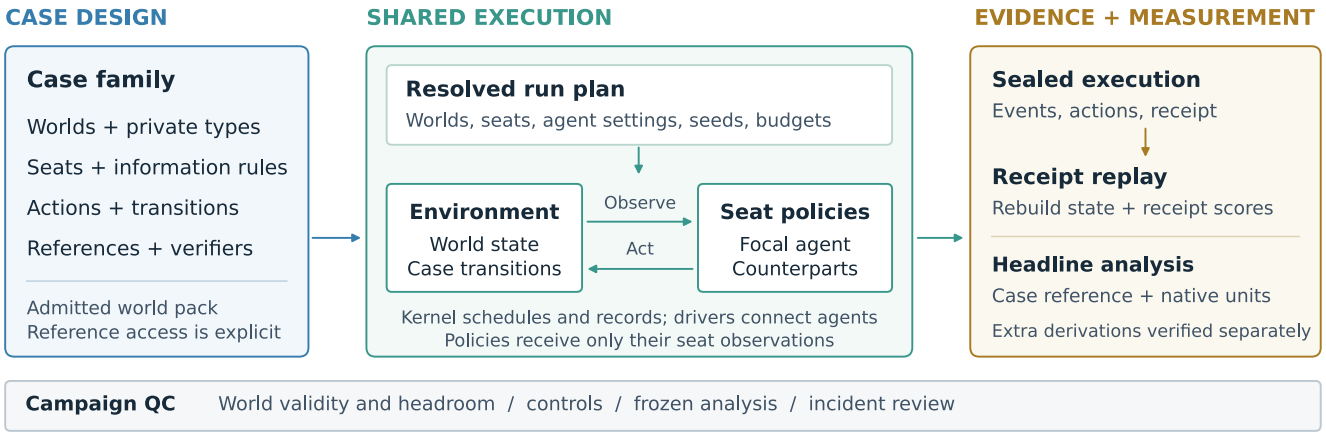


Figure 4.1. AERead architecture. Families define the economic world and reference; the runtime coordinates seat policies and retains execution evidence. Agents receive only their seat observations, while scoring uses explicitly declared reference access. Replay checks receipt consistency; separate headline derivations require their own verification. Applied coverage and QC verdicts vary by campaign (Appendices F and H).

4.1 Verification

A check compares an action or an outcome with a reference and reports a pass or fail, a count or an amount. The cases use four kinds of reference. Refund uses an answer key, which specifies the correct resolution of each claim. Procurement and the agreement case also check rules, which constrain what an action may do. Housing, Procurement and the data-center case measure value, comparing an economic outcome with a policy or a bound. The data-center case also compares named agents with each other. Results keep their native units, and we report no cross-family aggregate. A value reference is one of three types: a policy the evaluated seat could carry out, the best policy given the deciding seat's information, or a full-information upper bound that the seat need not be able to reach.

The case-study scores are computed by code. Kernel replay reconstructs the state and each receipt's score from the recorded actions, without calling a model. Replay therefore establishes that a score agrees with the recorded execution. Whether the score measures the intended decision is a separate question, which the quality-control checks of Section 6 address. Housing and Procurement also report headline measures that are computed outside the receipt and verified separately; Appendix H.3 documents how each is computed.

4.2 Evaluation harness

Before execution, the kernel hashes a plan that specifies the worlds, seat assignments, agent configurations, seeds and budgets. During simultaneous play, it freezes all acting seats' observations before requesting actions and then applies their joint transition. Calls through its provider and tool interfaces are logged with typed outcomes. Finalization seals the evidence and produces a receipt binding the plan, world and scores.

A seat is played in one of three ways. Most models run through AERead's own harness. Each action is a single call through a pinned provider route. The call receives the seat's instructions and its current observation, and it returns one action in the case's schema. It has no tools and no memory beyond what the observation shows. The harness checks the route against its pin, requests a seed, and records the specified reasoning setting and output limit. Gemini 3.8 Flash, GLM 5.3 Flash and GPT-6 Luna in Housing (Gemini and Luna in Procurement), the Refund models and the TERMS-Bench agents are run this way.

The two frontier agents, Opus 5.5 and Sol 6.1, run instead through their vendors' command-line tools, Claude Code and Codex. Each action starts a fresh process at low reasoning effort, with the same instructions and schema, tools disabled or refused, and no session carried over. Neither tool was fully isolated from the operator's machine. Claude Code ran in safe mode, with none of the operator's instructions, memory or skills, but its

working directory was in its context. Codex still listed eight of the operator's own skills (Table H.1). Only the tool's version and binary are checked, and the tool exposes no seed, temperature or output limit. Through the harness, the system under evaluation is the model; through a command-line tool, it is the product, including the tool's own instructions and internal turns. A comparison across the two paths therefore compares agent configurations rather than models, and each result names its path.

The third way is a script. Counterparts, references and simple-rule baselines are scripted, call no model and replay exactly.

A recorded setting does not establish that a provider honoured it. Appendix H gives the controls each path applied and the versions behind each reported panel.

5. Case studies

We present four cases (Table 5.1). Each is a starting point for questions about the decision it models, and Section 7.2 lists the questions we consider open. The cases show how the counterpart changes measured value and how a shortfall divides into typed action classes, and Section 5.3 runs an external benchmark on the same runtime. We describe Housing, Procurement and Data center here and specify them in Appendices A–D; Refund appears only in the cross-case comparison. Each case is a family rather than a fixed task set. It consists of a generator of worlds; a case-specific reference, which is either a policy that uses only what the deciding side may know or an explicitly labelled full-information upper bound; an inclusion criterion that keeps the worlds in which the decision matters (its form varies by case; Appendix E.2); and a set of verifiers. The references differ in what a shortfall means. Against a reachable policy, the shortfall is the value that a feasible strategy would have kept. Against the best policy given the decider's information, it is the whole cost of the decisions. Against a full-information bound, it also includes the value of information that no seat had (Appendix C).

We choose cases by the economic value of the decision rather than by occupation (Li 2026). An activity qualifies when its value is decided rather than produced, when a counterparty has its own interest and private information, and when the result can be scored within a world against a computed reference. The four examples lie in two of the five transaction classes introduced in §3, business and household purchases. We chose them to span the kinds of verifier and counterpart, not to sample a representative distribution.

Table 5.1. The four cases. Each case is scored in its own units against the reference given in the last column.

Case	Real setting	What the agent decides	Who it faces	Scored against
Housing	informal rental listings	which listing to inspect, what to offer, and whether to sign when some listings are lemons	scripted landlords; other tenants take listings it leaves	a scripted tenant that uses only what the tenant knows
Procurement	wholesale sourcing	pay more for an established supplier, or test a cheaper newcomer, before a deadline	scripted suppliers	the best plan given what the buyer knows, solved exactly
Data center	AI-infrastructure contracting	which risk clauses to change, what price to name, and when to sign or walk away	the other side: a script, another model or the same model	the best agreement at both sides' true costs, which neither side knows
Refund	customer refund claims	what to ask, decide and pay on a claim under a written policy	the other parties to the claim, scripted	a hand-written correct resolution

We report how much room each case leaves as a share of that case's own reference value. Tables 5.2 to 5.4 give each subject's result and gap in this form, and Figure 5.2 divides the gap by action class in the case's own units. The cases rely on four kinds of check: an answer key, stated rules, the value of the best policy or a bound, and relative standing against another agent. None uses a rubric. Every score is computed by code, with no model in

the loop. The kernel's replay recomputes each receipt's own score from its sealed events, whereas the headline measures of Housing and Procurement are computed by case code outside that certificate (Appendix F.2). Table E.1 lists every check.

5.1 Results

The four cases are worked examples that we built and maintain. Their results therefore compare the named agents on these worlds; they are not qualified benchmark scores. Each result is also bounded by its case's quality-control gates, which Section 6 describes and Table 6.1 summarises. Below, "pre-registered" describes how a comparison was run: its analysis was committed to the repository before any episode.

Housing. The structure follows informal rental markets on social platforms such as RedNote. Listings are posted with photos that a tenant cannot check before viewing, a viewing costs time, and good listings are often taken while the tenant decides. We set the rents, the lemon share and the inspection cost ourselves; they are not drawn from platform data. A tenant chooses among four listings, two of which are lemons. It knows its own value v_ℓ for each listing if sound, but not which two are lemons; each of the six pairs is equally likely. In each of three rounds it may pay \$25 to inspect one listing, make one offer, and sign or walk away from a lease it is offered, while each listing it did not bid on is taken by outside demand with probability ½. Signing listing ℓ at rent x after k inspections pays

$$v_\ell - 1{,}000 \cdot \mathbb{1}[\ell \text{ is a lemon}] - x - 25k, \tag{1}$$

and signing nothing pays -25k. The difficulty lies in information. A listing worth \$400 more than its ask if sound is worth $400 - \tfrac{1}{2}\cdot 1{,}000 = -\$100$ if signed blind at the ask, but $\tfrac{1}{2}\cdot 400 - 25 = +\$175$ if it is inspected first and signed only when sound; inspecting a second listing takes another round, in which each listing not bid on may be taken. A landlord's reply can reveal a lemon at no cost: every world is played against one landlord kind whose reply to a low offer reveals a lemon and one whose reply does not. We score a blind signing at its expected value given what the tenant knew, so that the score follows the decision rather than the draw, with the odds conditioned on every informative landlord reply. This paper introduces the corrected measure. The comparison reported here, v15, was pre-registered before any of its artifacts. Each included world is screened so that the best informed tenant beats a simple inspect-then-sign rule by at least \$25, and the tenant's task prompt states how landlords price. A holdout of 40 worlds changes one mechanism, outside demand at 0.3 a round, and its task prompt says so.

Table 5.2. Housing v15, main evaluation set (260 worlds; 200 with the favourite a lemon, 60 sound): each model's net payoff under the corrected measure as a share of the best informed tenant's (\$307 a world), averaged over the two landlord arms. Differences are in percentage points of that value, with 95% intervals over worlds.

	Gemini 3.8 Flash	GPT-6 Luna	Sol 6.1 (Codex), addendum	Opus 5.5 (Claude Code), addendum
Share of the best informed tenant's value	78.7%	62.9%	90.4%	80.4%
Favourite is a lemon / sound	68.8% / 100.0%	46.8% / 97.3%	85.9% / 100.0%	71.2% / 100.0%
Minus `inspect_then_sign` (the pre-registered beatability baseline), points	+29.3 [+24.7, +33.8]	+13.4 [+8.2, +18.6]	+40.9 [+36.2, +45.6]	+30.9 [+26.3, +35.5]
Minus `inspect_lowball` (v14's reachable reference), points	−16.6 [−20.8, −12.5]	−32.5 [−37.8, −27.3]	−5.0 [−8.0, −2.0]	−15.0 [−18.9, −11.0]
Holdout (40 worlds, outside demand at 0.3)	79.3%	61.1%	88.0%	80.1%

Both pre-registered models beat the pre-registered baseline and stay below the best informed tenant, and almost all of their shortfall arises in worlds whose favourite listing is a lemon. Gemini leads Luna by 15.9 points [11.3, 20.4] of the best informed tenant's value. This lead is significant under the pre-registered analysis. Its

lower end, however, falls below 16.3 points, the share that corresponds to the $50 we pre-registered as the smallest meaningful difference, so the lead is not shown to exceed that band. Where the favourite is a lemon, Gemini keeps 69% and Luna 47% of the best informed tenant's value; where it is sound, they keep 100% and 97%. Both models also stay below `inspect_lowball`, a simple rule that inspects the favourite and opens with the lowest possible offer, which earns 95% of the best informed tenant's value. The holdout points the same way (Table 5.2, last row).

Further runs. v15 differs from the earlier v14 comparison in four ways at once (the screened evaluation set, the stated pricing rule, the corrected measure as primary, and receipts that score the deciding tenant). Its levels are therefore not comparable with v14's, and we attribute no change between them to the models or to any single difference; Appendix E.4 reports v14. Two pre-registered addenda ran outside v15's contrasts. In the first, GLM 5.3 Flash, which was excluded at the pre-run profile check because its provider had changed quantization, ran on a provider that serves v14's quantization; it does not beat `inspect_then_sign`. In the second, Sol 6.1 and Opus 5.5 played through their command-line tools. Sol keeps the largest share of any model, and Opus cannot be distinguished from Gemini. Appendix E.4 gives the addenda's contrasts, the A/A arm that calibrates the $50 band, and the holdout.

Procurement. The structure follows wholesale sourcing on B2B platforms such as Alibaba's 1688. A displayed price is not a quote, a listing's quality is known only after it is checked, and new sellers of one kind often buy from the same factory. A buyer owes 50 kits by a deadline and awards the order to one supplier: either a recorded supplier with a long clean record or one of three newcomers. Each newcomer is good, late or poor. The buyer knows only the odds, which are read from the record of the newcomer's seller kind, and that newcomers of one kind share a source with a stated probability (0.5 to 0.95), in which case they are all of the same type. Two features follow our own sourcing on 1688. A newcomer's price relative to the recorded supplier's is drawn from the price gaps in that record, so a newcomer is dearer about half the time, and a newcomer answers a quote request only a third of the time. We set the case's other quantities ourselves. Among up to ten actions the buyer may request a quote ($2 and a day; an answered quote exposes a late supplier), request a sample ($20 to $240 and three days; it always arrives and exposes a poor supplier), look up a listing's registered factory ($1; two lookups in one kind show whether the kind shares a source), award the order, or defer. An award that delivers k kits on time earns

$$37k - 12(50 - k) - \text{cost of the order} - \text{cost of the information bought}, \tag{2}$$

and an award that delivers fewer than 40 kits, as a late or poor supplier does, earns −$600 less the information bought (Appendix A.2).

The difficulty is a test worth taking only for what it reveals about another supplier. Every world in the evaluation set is one in which a rule that plans a single newcomer's test exactly (what to request first, how often to repeat an unanswered quote, and when to give up) pays the premium, while the best policy tests a newcomer first and beats paying the premium by at least 10%. In one world the recorded supplier asks $24.95. A newcomer asks $12.47, but its sample costs $200 and its odds are 0.20 good, 0.15 late and 0.65 poor, so testing it on its own loses money. A second newcomer of the same seller kind asks $26.87, more than the recorded supplier, so it is never worth buying; but its sample costs $30, and the two share a source with probability 0.95. Once the cheap newcomer's quote has come back clean, a sound sample of the dear newcomer raises the cheap newcomer's probability of being good to 0.94, and a poor sample lowers it to 0.01. The best policy therefore requests a quote from the cheap newcomer; if the quote comes back clean, it samples the dear newcomer, looks both up, and only then pays for the $200 sample. It earns 11.8% more than paying the premium. Every action is charged the value it gave up relative to the exact best informed policy at that point.

Table 5.3. Procurement, the 20 pre-registered worlds, two plays each: the share of the gap between paying the premium and the best policy that each model closes (a mean over worlds, with 95% t intervals), the loss per world relative to the best policy, and the plays that touched a newcomer.

	Share of the gap closed	Loss per world	Plays touching a newcomer
Sol 6.1 (Codex)	35% [18, 52], p = 0.0003	$32.51	26 of 40
Opus 5.5 (Claude Code)	0% in every world	$51.25	0 of 40
Gemini 3.8 Flash	0% in every world	$51.25	0 of 40
GPT-6 Luna	−14% [−21, −7]	$58.10	0 of 40
Exact rules on the same worlds			
Paying the premium; a rule that plans one newcomer's test	0%	$51.25	
Shortcut: sample the newcomer with the cheapest sample first, ignoring seller kinds	33%	$33.46	
A rule that may test two newcomers	82%	$8.38	

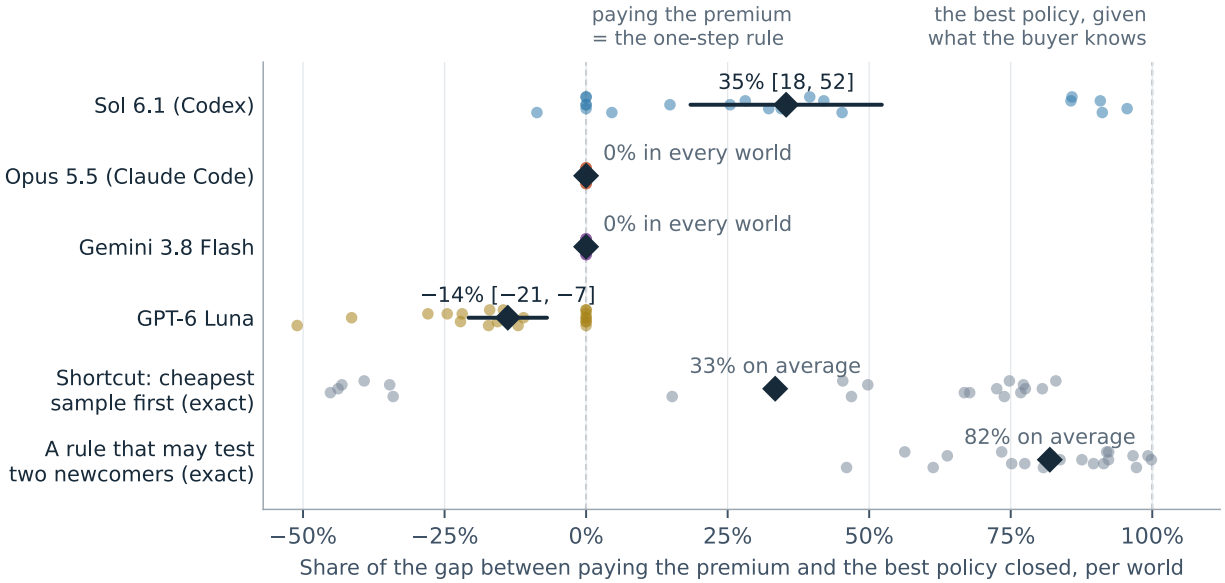


Figure 5.1. Procurement, the 20 pre-registered worlds: the share of the gap between paying the premium and the best policy closed, world by world. Each dot is one world, the mean of its two plays; diamonds are means, bars are 95% t intervals, and the gray rows are exact rules.

Sol 6.1 closes 35% of the gap (95% t interval 18–52%, p = 0.0003), whereas a rule that plans one newcomer's test exactly closes none. The worlds were drawn, and this test pre-registered, before any play, and the claim concerns Sol alone. Sol's play varies by world (Figure 5.1): it closes most of the gap in some worlds and none in others. The other three models never touched a newcomer and paid the premium. Luna falls below paying the premium because it ordered 50 units where 55 was best in 15 of its 40 plays. A shortcut that ignores seller kinds and first samples the newcomer whose sample is cheapest closes 33%, about Sol's share. Sol does not play that shortcut, however; its tests follow price. A rule allowed to test two newcomers closes 82%, so the remaining headroom lies in planning one test beyond the one in hand.

Further runs. A development pilot and four diagnostics on this evaluation set, among them a frozen control, leave this picture unchanged (Appendix E.4, Table E.15). One of them bears on why Opus never tests. Asked, with no play, how a sample of one newcomer changes the odds of another of its kind, Opus answered all 18

questions within 0.05 of the exact value, as Sol did. Opus therefore applies the sharing rule when asked, but does not plan with it. On an earlier evaluation set, in which quotes always arrive and newcomers never share a source, a rule that looks one test ahead comes within 0.01% of the best policy, so that set cannot separate the models (Appendix E.4).

Data center. The structure follows a real AI-infrastructure engagement in which a system integrator bundles hardware, data-center space, staffing and software into a working system for a client. Hardware can arrive late or defective, the site may not be ready, and the contract's warranty, delay-damages, liability-cap, site-readiness, deposit and burn-in clauses decide who bears each cost. We set the costs, probabilities and prices ourselves. Client and integrator negotiate one agreement, consisting of a position on each of the eight clauses (600 distinct agreements) and an all-in price p. Each side knows its own cost of carrying risk and only the probability distribution of the other's: the integrator's pre-staging cost and risk charge, and the client's risk charge. Over six alternating moves, a side may redline any clauses and name a price, sign the standing version, or walk away, and each redline after the first move costs $25k. In thousands of dollars and net of redline fees, a signed agreement c pays the integrator and the client, respectively,

$$\pi_I = p - C_I(c) - 500, \pi_C = O_C - p - C_C(c), \tag{3}$$

where 500 is the integrator's outside option, O_C the client's, and C_I and C_C each side's expected cost of the risks that the clauses leave with it, marked up by its risk charge; no deal pays only the fees. The difficulty is that price only divides the value, while the clauses determine how much value there is, and the best clauses depend on what the other side hides. Burn-in, for example, costs the integrator $40k and, in one world, costs the client a week's delay worth $110k, and it halves the probability of a $3 million incident that the client carries. It therefore lowers joint cost by $41k for a client with the highest risk charge and raises it by $19k or $64k for the other two, and the integrator cannot tell which client it faces. The comparison reported here (v2) was pre-registered before its runs, on 80 included worlds (from 5,436 scanned), 72 of them with a mutually beneficial deal. Both command-line tools were isolated, the client's task prompt states its walk-away point, and a world is included only if the integrator has at least $25k of room over a price-only rule.

Table 5.4. Data center v2, the pre-registered comparison of Sol 6.1 (Codex) and Opus 5.5 (Claude Code) on the 72 deal worlds: differences in a seat's earnings above its outside option, at true costs, in percentage points of the value available in a deal ($309k a world), with 95% intervals and Holm correction across the four primary contrasts.

Primary contrast	Worlds	Estimate [95% interval]	Holm p
Head to head: Sol minus Opus earnings, seat-averaged	72 deal	+41.7 [31.3, 52.0]	6.1e-11
Integrator against the firm client: Sol minus Opus	72 deal	+7.5 [1.7, 13.2]	0.022
Sol as integrator minus the price-only rule	72 deal	+7.3 [2.6, 12.0]	0.008
Opus as integrator minus the price-only rule	72 deal	−0.2 [−5.5, 5.2]	0.95

The counterpart decides most of the measured gap. Head to head, Sol's earnings exceed Opus's by 41.7 points of the value available, but against the same scripted firm client the gap is 7.5 points (Table 5.4). Both gaps are significant under the Holm correction. Neither is shown to reach its pre-registered minimum, because the lower ends of their intervals fall below 32.4 and 12.9 points respectively ($100k and $40k). Head to head, the opponent changes with the focal model, whereas the firm-client contrast holds the counterpart fixed, so most of the larger gap is specific to the counterpart. As integrator, Sol beats the price-only rule, and Opus earns what the rule earns. As client, with its walk-away point stated, neither model signs below that point, and the client seat separates nothing. After both runs we computed the best informed turn-by-turn policy against the same scripted client (Appendix C.3). Sol captures about 20% of the room between the price-only rule and that policy. Knowing the client's type is worth less than 1% of the value available to that policy, so the case measures bargaining against a fixed counterpart far more than inference about it.

Further runs. An earlier pre-registered comparison (v1) ran without isolated command-line tools, without the client's walk-away point in its task prompt, and without the inclusion screen against the price-only rule. Appendix E.4 reports it, with the development results and the figures behind this paragraph.

5.2 Where the shortfall comes from

Figure 5.2 divides each case's shortfall into observed action classes. The components sum to the total under a stated priority order, and they describe what happened, not its cause. The Housing panel shows v15's two pre-registered models and three addendum agents (A: GLM; B: Sol and Opus through their command-line tools), measured below the reachable rule `inspect_lowball`. Negative components, at most $0.4 a world, are not drawn. The Data center panel shows v2's joint loss relative to the full-information bound against the firm counterparts, with each redline after the first move charged $25k, across all 80 worlds with both seats averaged. Appendix E.5 gives the classification rules and full subject tables, Appendix E.7 the Data-center breakdown and its uncertainty, and Tables 5.3, E.9 and E.10 the intervals and exact values.

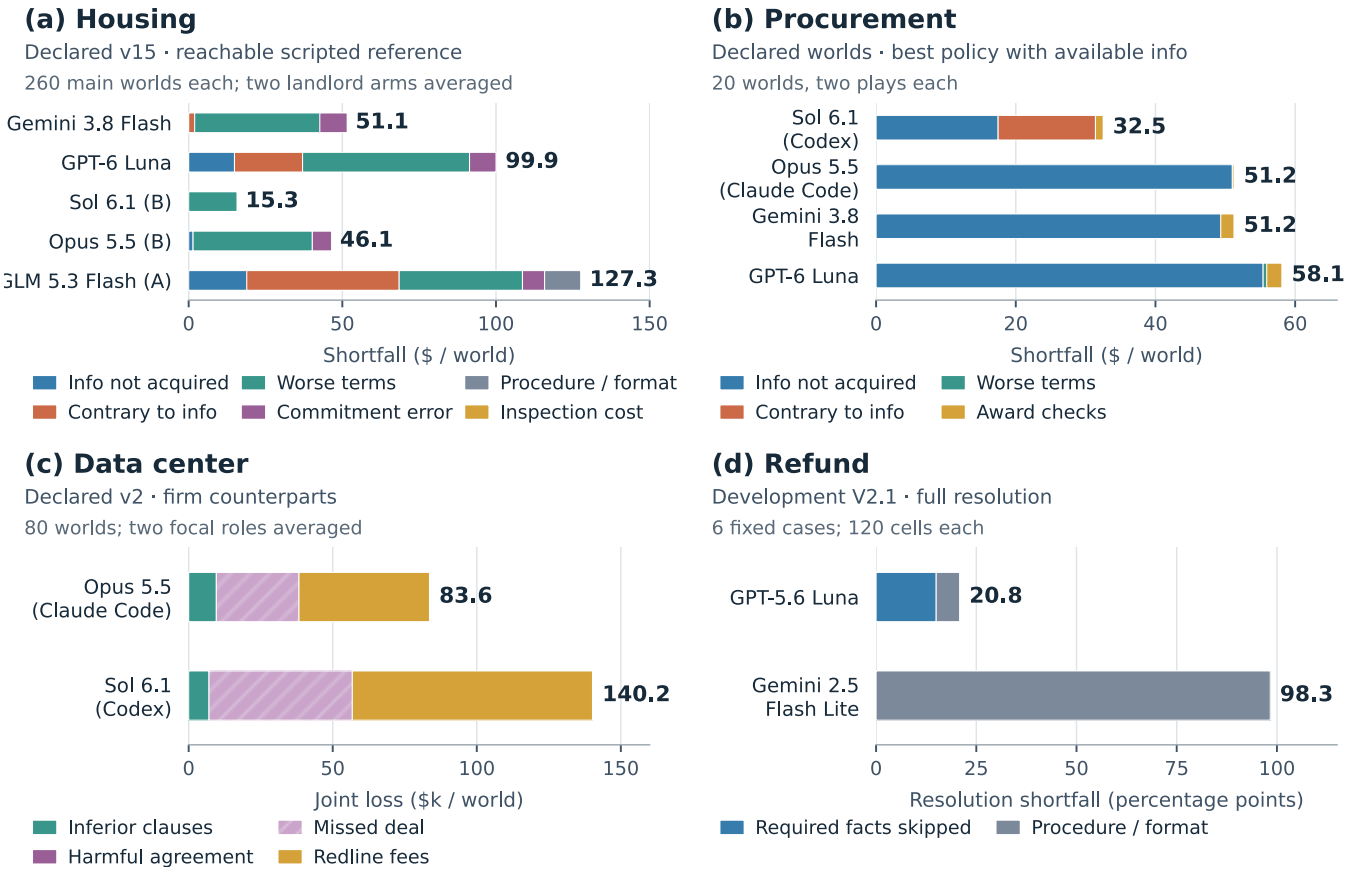


Figure 5.2. Shortfall to each case's reference, divided by observed action class. Labels give mean totals (lower is better); each panel has its own reference, units and scale. Housing uses the 260 main worlds of Table 5.2, Procurement the 20 worlds of Table 5.3, and Data center the 80 worlds of Table 5.4.

The labels name what happened. The mechanism behind each large segment is as follows.

Procurement: three models never test, and Sol secures the recorded supplier first. Opus, Gemini and Luna asked no newcomer for anything in any of their 120 plays: they requested quotes and samples from the recorded supplier and awarded it the order, so nearly all of their loss is information not acquired, and the rest is award checks, each of which spends an action. Luna's further $6.85 a world comes from ordering 50 units where 55 was best in 15 of its 40 plays. The pre-registered classes file an award's whole loss under information not acquired when the award came before a worthwhile test, so $6.33 a world of that worse-quantity loss sits

in Luna's information-not-acquired segment (Table E.10). Sol's loss divides by world. Where it never tested a newcomer that the best policy would have tested, it lost $17.5 a world. Where it did test, it often first requested a quote and a sample from the recorded supplier, spending days and actions before the newcomer test that the stated odds and clock made best ($8.9 a world), or it awarded the order while a newcomer test it had begun was still worth finishing ($3.2).

Housing, Gemini: it drops a known lemon even where the lemon is the better lease. Worse terms account for 80% of its shortfall relative to the reachable rule ($40.7 of $51.1 a world), and nearly all of it comes from one move. Under the landlord kind that prices a lemon from its own lower cost, a lemon's landlord counters about $1,000 below a sound listing's floor ($995 to $1,000 in these worlds), so a known lemon signed at that counter is worth what the listing would be worth sound at a sound floor. This holds exactly in 187 of the 200 worlds whose favourite is a lemon, and within $5 in the rest. The rule inspects the favourite, finds the lemon, offers $0 and signs the counter in all 200. Gemini also inspected the favourite and found the lemon in all 200, but made an offer on it in only 4; in 172 episodes it leased a sound listing that left it with less ($65.0 a world). Under the other landlord kind the rule leases nothing in 77 worlds, and Gemini's leases there offset $26.5 a world. Gemini does read landlords' replies: of 74 replies that proved a lemon, it walked away from 64 and signed above the lemon's value once. Ending with no lease where the rule leased costs a further $8.9 (12 of 520 episodes).

Housing, Luna: it drops the lemons worth taking and signs ones that cost it. Its shortfall relative to the reachable rule is $99.9 a world. Worse terms (54%, $54.3) follow Gemini's pattern: in the same 200 worlds Luna found the lemon favourite in 197 and made an offer on it in 7, and in 154 episodes it leased a sound listing that left it with less ($57.7 a world). Signing a listing that a landlord's reply had proved a lemon accounts for 22% ($22.2, 11 episodes): the landlord accepted Luna's offer at a rent that no sound landlord would take, which by the stated pricing rule proves a lemon, and Luna signed at that rent, above the lemon's value. Of 89 such replies, it walked away from 58 and signed 20 at or below the lemon's value. Another 15% ($14.9, 64 episodes) comes from signing without settling quality under the landlord kind whose replies reveal nothing, where an inspection was worth its $25 fee every time; 21 of those listings were lemons. Ending with no lease where the rule leased costs $8.6. In v14, whose task prompt did not state how landlords price, the reply-proved lemon accounted for 60% of Luna's shortfall.

Housing, Sol and Opus (addendum B): what remains is the price. Neither lost anything by acting against what it knew, and neither left a worthwhile inspection undone, apart from Opus's $1.4 a world. Worse terms account for all of Sol's $15.3 shortfall ($15.7, offset by $0.4 of inspection fees it saved) and for 84% of Opus's $46.1 ($38.8); ending with no lease where the rule leased costs Opus a further $6.3.

Refund, Gemini 2.5 Flash Lite: the case ends before it decides. In 81% of its episodes it asked again for a fact it had already been given; the environment rejects a repeated request, and the case ends with no decision (80.8 of its 98.3 points). In most of the remaining episodes it denied the claim correctly but wrote the refund method as null where the policy's format requires "none" (16.7 points). It reached the answer key's outcome in 18% of episodes and a wrong outcome in none.

Data center: fees, then deals the client walked away from. Every redline after the opening move costs $25k. The full-information reference signs the best agreement at once, so each fee counts as loss, whichever side paid it. Against the firm counterparts, across all 80 worlds with both seats averaged, fees account for 59% of Sol's joint loss ($83.4k of $140.2k a world) and 54% of Opus's ($45.3k of $83.6k). Sol redlines more: as integrator it made 371 redlines to the firm client's 262, whereas Opus made 227 to 189. Some of those redlines buy better clauses, and a seat that does not know the other side's costs may need them, so their share alone does not show wasted negotiation. Nearly all of the remaining loss is deals lost in worlds that had one (35% of Sol's loss and 34% of Opus's), and every one of them ended with the client walking away. With each model as integrator,

the firm client walked away from Sol in 40 of its 144 plays of deal worlds and from Opus in 14. As client, each model walked away in 26 of the 72 deal worlds. No agreement was harmful: with the walk-away point in the task prompt, neither model signed in any of the 8 worlds without a deal.

The pre-registered plan included a replay of the recorded moves at other fees, which shows what the same moves would earn, not how the models would play. Without fees, Sol still loses more joint value than Opus against the firm counterparts ($63.1k against $42.5k a world, seats averaged) and earns more in both seats. At $50k a redline its earnings fall below Opus's (as integrator 67.9 against 69.7, as client 1.3 against 22.8, in thousands of dollars per deal world), because it redlines more. Appendix E.9 gives v1's fuller replay.

Across the cases. Four findings hold across the cases. First, the gap lies in a stratum that the design fixes in advance, so every case reports by stratum and pools last. Second, the counterpart is part of the measurement, so a claim names the counterparts it covers. Third, what the task prompt states moves the score as much as the subject does (stating two rules cut procurement loss by $11 to $14; stating a walk-away cost cut losing signatures from 15 of 60 to 1 of 60), so the task prompt is a versioned part of the case. Fourth, a pattern found by looking back is a hypothesis until a pre-registered test confirms it.

5.3 An external benchmark on the same runtime

TERMS-Bench evaluates bilateral price negotiation against counterparts whose type is hidden (Zhang et al. 2026). We reimplemented, from the paper's description, its alternating-offer environment, three of its six counterpart families (Candid, Taciturn and Expressive) and four of its measures as one AERead family. The counterpart runs as a scripted seat, and its random draws are sealed each turn, so replay recomputes the counterpart's moves as well as the scores. We did not implement the Oracle-Cue reference or the gap measures built on it, the urgency regime, type-belief error or the other three families, and counterpart messages are templates. The corpus has 30 cases, 15 with a zone of agreement and 15 without, each played once.

Table 5.5. TERMS-Bench reimplementation: two of the paper's agents on the same 30 cases against scripted counterparts, beside the paper's values. Agreement and surplus are computed over the cases with a zone of agreement (15 for GPT-4o-mini, 12 for GLM-5.1).

	GPT-4o-mini	GLM-5.1	Paper: GPT-4o-mini	Paper: GLM-5.1
Agreement rate where a deal exists (AGR+)	0.80	1.00	0.522	0.951
Surplus efficiency (SE+)	0.035	0.458	0.196	0.668
Conditional surplus efficiency (CSE+)	0.044	0.458	0.363	0.721

The paper's surplus efficiency comes from its Table 16 (overlap regime, mean of the three families played here); its agreement and conditional surplus efficiency come from its Table 2, which pools six families and three regimes, so those two comparisons are looser. GLM-5.1 has 12 cases because its three malformed moves removed those episodes.

The reimplementation orders the two agents as the paper does on all three measures, but absolute levels are lower for both, and for a third model. A review of the port after these runs traced the shift to four departures from the paper's protocol rather than to the agent, whose interface in the paper is also one stateless call per round. First, our prompt does not state the agent's payoff or the rule against accepting above its reservation value. Second, its observation differs from the paper's. Third, a malformed move removes the episode where the paper applies a fallback; under that fallback, GLM-5.1's surplus efficiency would be 0.520 over all 15 cases. Fourth, our price range and zone widths are wider than the paper's.

Further runs. We have built a pre-registered reproduction with those fixes, and it stopped at its first check: one of the paper's fixed-concession baselines (FC10) scores 0.307 here against the paper's 0.266, while the other two match. The paper states how much the baselines concede (1%, 10% or 30% of the remaining distance per offer) but not how they open or accept. No agent has played the reproduction. The paper reports critical violations by GLM-5.1 in 5% of cases without a zone of agreement for these three families; at that rate, 15 such cases would show none about half the time, and ours showed none. The port also exposed a harness effect. On one provider route, any specified reasoning budget suppressed deliberation, and GLM 5.3 Flash breached its own reservation value in 13 of 30 cases, against the paper's band of 0 to 2.06%. With no reasoning setting specified, it breached in 2 of 30 and its surplus efficiency rose from 0.40 to 0.53; the comparison above uses that setting. This is why the reasoning condition is part of each run's frozen configuration. The runtime's support for the scripted seat that this family uses is now on the main branch.

6. Quality control

The defects that mattered in the four cases were properties of the distribution, the task prompt, the counterpart or the analysis, and every world was individually correct when they were found. In one, a seat-number tie-break decided who signed the good listing; in another, a prompt pointed every agent at the right supplier in half the worlds and at the wrong one in the other half; in a third, a hidden type was tied to the stratum of the evaluation set, so that comparisons by type compared worlds. An inspection of one world at a time passes all three. We therefore organise quality control around populations of worlds, pre-registered analyses and recorded failures. Appendix F describes each check that we applied to these cases, and only those.

Gates. A case is qualified through six quality-control gates, whose verdicts are recorded in its family's QC profile. Gate 0 requires that the profile exists; Gate 1, that the worlds are valid, distinct, split and leave headroom; Gate 2, that actions change the intended state and that every score is rebuilt from sealed evidence; Gate 3, that a better-informed policy does better and that shortcuts lose; Gate 4, that the result supports the attribution it claims; and Gate 5, that the comparison was sized, frozen and executed without tuning on outcomes. Each gate has one of five statuses (`passed`, `failed`, `partial`, `not_run` or `not_applicable`), and only `passed` qualifies a result. We wrote the profiles after the initial runs. The confirmatory run of the agreement case came after its profile. Housing and Procurement pass four of Gates 1 to 5, all except Gate 4, and no case yet passes all five (Table 6.1).

Table 6.1. Gate status of each reported comparison on 21 October: ● passed, ◐ partial, ✗ failed, – not run. Appendix F.1 gives the reasons and the flags each check raised.

Comparison	0	1	2	3	4	5	Main open item
Housing v15	●	●	●	●	◐	●	profile probe ran after the freeze
Procurement, pre-registered evaluation set	●	●	●	●	◐	●	the agents take no seed
Data-center agreement v2, integrator seat	◐	◐	◐	●	◐	◐	not on the main branch; Gate 1 needs a new evaluation set
Refund V2.5, developmental	●	◐	◐	✗	◐	–	a constant full refund passes 38 of 180 cases

Headroom (Gate 1). When a case computes its reference exactly, as Housing and Procurement do, every shortcut named in its profile must lose a pre-specified margin to the reference in every world. Otherwise, a frozen model control screens the worlds. We added the first route on 21 October (Appendix F.1). The data-center agreement's reference is a lower bound, and its evaluation set is frozen, so its Gate 1 can pass only on a new evaluation set.

The reference is played on every world. For each case, a test plays the reference through the real environment and verifier on every included world, with no model, and checks that it reaches its stated result. Every reported evaluation set has such a test; Appendix F.2 lists them, including the ones we wrote after the runs they check. Two of the tests, the data-center grade and the Housing ceiling, are also shown to fail when the grade or the reference is wrong.

Other checks. Every committed evaluation set is regenerated byte for byte from its seeds, and the Housing v14 and v15 receipts replay from their sealed events. We report trivial agents beside the evaluated agents, score Procurement's fixed rules exactly, and audited the task prompts for instructions that decide the outcome. Uncertainty is computed over worlds, and every claim names its counterparts. Six reported analyses were pre-registered: Housing v14 and v15 (with v15's two addenda), Procurement's main test, the data-center signalling test and the two confirmatory runs of the agreement case. Procurement's frozen-control diagnostic was also pre-registered before its plays.

What it found. We record every discovered failure in an incident log, as design, operational, tooling or judgment rows that are never deleted. On 16 October the log held 51 entries about Housing, 57 about Procurement, 90 about the data center and 23 about Refund (Table F.3), and rows have been added since. Of the 25 defects selected for Table F.4, 14 were invisible in any single world. We found the 11 visible ones by reading code or sealed bytes, by tracing a case, or within a single gate. Nine defects were open when we compiled the table. A later reanalysis from the sealed actions resolves one of them for the reported panels, which leaves eight open. The same checks corrected three of our own statements while we wrote this paper.

---

7. Discussion and conclusion

7.1 What the cases show

AERead evaluates an agent’s economic decisions by the value it obtains for a principal, given the information, resources and authority of its seat. Each score is measured against a reference whose access to information is stated. It keeps its own units and names the counterparts it was measured against, so every shortfall has a defined interpretation. Our cases apply this to information acquisition, bargaining and commitment in housing, procurement and contracting, and a reimplementation of part of TERMS-Bench runs on the same runtime.

The cases teach three lessons that bear on any evaluation of this kind. The counterpart is part of the measurement: the same two agents differ by far more when they negotiate with each other than when each faces a fixed counterpart. The task prompt is part of the case: stating a rule that the prompt had left implicit changed scores as much as changing the agent did. And the defects that matter are often properties of the population of worlds, the task prompt or the analysis rather than of any single world. Construction and quality control therefore belong in the method, and the checks of Section 6 changed the interpretation of our own results. Loss categories locate a shortfall but do not explain it, and explaining it requires controlled follow-up comparisons. Replay establishes consistency with recorded play; construct validity requires evidence about the worlds, the information and the controls.

These properties matter most where agents are deployed. Organizations that pilot agents often stop because the business value is unclear and the risks are hard to control (Gartner 2025); a value score against a stated reference and a typed account of losses address both. Small firms face a different problem: short of staff (National Federation of Independent Business 2026), they often leave decisions like those our cases model to one person, without specialized expertise and with no review before a commitment. Delegating such a decision to an agent is attractive there, and trusting it requires evidence that the agent keeps value against the counterparts it will meet.

The present evidence has clear limits. The cases take their decision structure from informal rentals, wholesale sourcing and AI-infrastructure contracting, but we set their parameters; calibration to field data and transfer to field settings remain to be established. Coverage is limited to two transaction classes and a small set of agent configurations, with no human policy baseline. No case passes all of the quality-control gates (Section 6). Several checks were added after the runs they qualify. The behavior of hosted models and command-line tools also limits how far a result can be attributed to its stated settings. Versioned reanalyses now regenerate the corrected Housing measures and Procurement decision losses from sealed records, which strengthens their provenance. Full verification against the original execution still requires the retained logs (Appendix G). The reanalyses neither qualify the campaigns retroactively nor resolve Housing's historical seat attribution (Appendix H). The results therefore do not establish general model rankings or deployment readiness.

7.2 Open research directions and contributions

AERead is shared infrastructure for studying how agents make economic decisions. Its code is open source under the Apache-2.0 license (github.com/aeread-org/AERead; contact@aeread.org), and contributors can pursue their own questions with the existing cases, counterparts and evidence. The results above leave four questions open, each suited to a different kind of contribution.

- **Counterpart populations.** In the Data-center case, the gap between two agents depended strongly on whether they negotiated against each other or each faced a fixed script. How far does an agent's measured skill depend on whom it faces, can rankings reverse across counterpart populations (none did in our runs), and which populations separate distinct strategic capabilities? The scripted styles, head-to-head pairings and counterpart seats are in place.
- **Validity of populations of worlds.** More than half of the defects we selected from our record were invisible in any single world. Which checks find such defects before a campaign runs, and what should qualify a case? The quality-control gates, the incident record and the failed campaigns are available for study.
- **Information and judgment.** Stating a rule that the task prompt had left implicit changed scores as much as changing the agent. Separating what an agent was told from how it decided, and computing best-informed references for richer cases, are open methodological problems.
- **New economic decisions.** A case for another activity supplies its rules, worlds and reference; execution, replay, counterpart seats and the quality-control standard come with the runtime.

Some components are ready and others are not. The runtime, its sealed evidence and replay, and the case layout are in use, and Housing, Procurement and Data center each have a world generator, references and scripted counterparts. No case is yet fully qualified. Section 6 gives every gate verdict, and Appendix F lists each campaign's open checks, so a contributor can see what a case still needs before its results can rank agents.

The next steps follow from the limits in Section 7.1. Future campaigns should bind the actual seat assignments and headline scorers into frozen evaluation configurations and run the required controls before any comparison, and the TERMS-Bench reimplementation needs the protocol fixes described in Section 5.3 before a preregistered reproduction. Practitioner review would calibrate the cases' parameters, human baselines would

show where agents stand against people, and externally contributed cases would test whether the structure serves authors other than its maintainers. AERead's present contribution is that structure, a common way to construct evaluations of economic decisions, together with an evidence-based account of the checks needed to interpret what they measure.

References

Barres, Victor, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. 2025. "τ²-Bench: Evaluating Conversational Agents in a Dual-Control Environment." *arXiv Preprint arXiv:2506.07982*. https://arxiv.org/abs/2506.07982.

Bianchi, Federico, Patrick John Chia, Mert Yuksekgonul, Jacopo Tagliabue, Dan Jurafsky, and James Zou. 2024. "How Well Can LLMs Negotiate? NegotiationArena Platform and Analysis." *arXiv Preprint arXiv:2402.05863*. https://arxiv.org/abs/2402.05863.

Feng, Yeqi, Yuxin Chen, and Tianxing He. 2026. "SidConArena: An Environment Evaluating Agents in Open-Ended, Positive-Sum Bargaining Game." *arXiv Preprint arXiv:2606.27397*. https://arxiv.org/abs/2606.27397.

Fish, Sara, Julia Shephard, Minkai Li, Ran I. Shorrer, and Yannai A. Gonczarowski. 2025. "EconEvals: Benchmarks and Litmus Tests for Economic Decision-Making by LLM Agents." *arXiv Preprint arXiv:2503.18825*. https://arxiv.org/abs/2503.18825.

Gartner. 2025. "Gartner Predicts over 40% of Agentic AI Projects Will Be Canceled by End of 2027." https://www.gartner.com/en/newsroom/press-releases/2025-06-25-gartner-predicts-over-40-percent-of-agentic-ai-projects-will-be-canceled-by-end-of-2027.

He, Horace, and Thinking Machines Lab. 2025. "Defeating Nondeterminism in LLM Inference." https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/.

Lanctot, Marc, Edward Lockhart, Jean-Baptiste Lespiau, Vinicius Zambaldi, Satyaki Upadhyay, Julien Pérolat, Sriram Srinivasan, et al. 2019. "OpenSpiel: A Framework for Reinforcement Learning in Games." *arXiv Preprint arXiv:1908.09453*. https://arxiv.org/abs/1908.09453.

Li, Chenyu. 2026. "Activities, Not Occupations." https://aeread.org/blog/activities-not-occupations.html.

Liu, Xianyang, Shangding Gu, and Dawn Song. 2026. "AgenticPay: A Multi-Agent LLM Negotiation System for Buyer-Seller Transactions." *arXiv Preprint arXiv:2602.06008*. https://arxiv.org/abs/2602.06008.

Merrill, Mike A., Alexander G. Shaw, Nicholas Carlini, Boxuan Li, Harsh Raj, Ivan Bercovich, et al. 2026. "Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces." *arXiv Preprint arXiv:2601.11868*. https://arxiv.org/abs/2601.11868.

National Federation of Independent Business. 2026. "NFIB Jobs Report: Small Business Hiring Slows in September." https://www.nfib.com/news/press-release/nfib-jobs-report-small-business-hiring-slows-in-september/.

Patwardhan, Tejal, Rachel Dias, Elizabeth Proehl, Grace Kim, Michele Wang, Olivia Watkins, Simón Posada Fishman, et al. 2025. "GDPval: Evaluating AI Model Performance on Real-World Economically Valuable Tasks." *arXiv Preprint arXiv:2510.04374*. https://arxiv.org/abs/2510.04374.

Sun, Yiyou, Xinyang Han, Weichen Zhang, Yuanbo Pang, Tianyu Wang, Yuhan Cao, Yixiao Huang, et al. 2026. "Agents' Last Exam." *arXiv Preprint arXiv:2606.05405*. https://arxiv.org/abs/2606.05405.

Terry, J. K., Benjamin Black, Nathaniel Grammel, Mario Jayakumar, Ananth Hari, Ryan Sullivan, Luis S. Santos, et al. 2021. "PettingZoo: Gym for Multi-Agent Reinforcement Learning." In *Advances in Neural Information Processing Systems*. Vol. 34. https://arxiv.org/abs/2009.14471.

Trivedi, Harsh, Tushar Khot, Mareike Hartmann, Ruskin Manku, Vinty Dong, Edward Li, Shashank Gupta, Ashish Sabharwal, and Niranjan Balasubramanian. 2024. "AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents." *arXiv Preprint arXiv:2407.18901*. https://arxiv.org/abs/2407.18901.

Zhang, Erica, Fangzhao Zhang, Aneesh Pappu, Batu El, Jose Blanchet, Susan Athey, Jiashuo Liu, and James Zou. 2026. "TERMS-Bench: Diagnosing LLM Negotiation Agents Beyond Deal Rate." *arXiv Preprint arXiv:2605.13909*. https://arxiv.org/abs/2605.13909.

Zhu, Yuxuan et al. 2025. "Establishing Best Practices for Building Rigorous Agentic Benchmarks." https://arxiv.org/abs/2507.02825.

Appendix A. Case specifications

Following TERMS-Bench (Zhang et al. 2026), we write each case as an environment Γ (outcomes, private types and their prior, payoffs), a protocol (seats, order of play, actions, what each seat observes, termination), and fixed counterpart policies (Appendix B). We do not specify Refund here. Symbols are local to each case, and each case's table lists them together with the kind of quantity each one is. The subscript A denotes the evaluated seat and B a counterpart. In controlled comparisons, the prior and the counterpart policies are fixed across evaluated agents; head-to-head role swaps change the opponent as well, so they compare pairings rather than agents. Agent configurations also differ in the settings recorded in Appendix H. We compute the reference policies of Appendix C in the same Γ.

Notation used across cases. $\mathbb{1}[\cdot]$ (or $\mathbb{1}(\cdot)$) is 1 when the condition inside holds and 0 otherwise. U(a, b) is a uniform draw between a and b. $\mathbb{E}$ is an expectation over the outcomes the case lists, and $\lfloor x \rfloor$ rounds x down. D denotes each case's loss: the lemon loss ($1,000) in Housing and the value of supplying nothing (−$600) in Procurement. Each case's table defines its other symbols, and Appendices C and D refer back to it when they reuse one.

Terms used across the cases. A *world* is one fully specified environment instance of a case, that is, its parameters and hidden types, fixed by a seed. An evaluation set is a published set of worlds that can be regenerated from their seeds, and its inclusion criterion, computed before any agent plays, decides which generated worlds enter it. A *seat* is a player position (the tenant, a landlord, the buyer, the integrator); the evaluated agent holds one seat, and scripted policies or other agents hold the rest. An episode is one play of one world under one condition (in Housing, one world and one landlord kind). A frozen configuration (such as v14 or v15) fixes an evaluation set, a task prompt, the counterparts and the controls; changing any of them creates a new configuration. A comparison is pre-registered when its plan and analysis were committed to the repository before any of its episodes ran. A *reference* is the policy or bound against which a case measures a shortfall, with its access to information stated (Appendix C). A *gate* is one of the six quality-control gates of Section 6, numbered 0 to 5: a check that a result must pass before it is reported as qualified.

A.1 Housing: lemons with outside demand (frozen configurations v14 and v15)

The v15 configuration uses the same rules as v14. Its task prompt states the landlords' pricing rule, its worlds come from a screened evaluation set (Appendix E.2), and its holdout sets the outside-demand rate to 0.3 a round.

Environment. A world seed w fixes four listings $\ell \in \{0, 1, 2, 3\}$ and six tenant seats. Seat 0 is the evaluated tenant, seat $i \in \{1, \ldots, 4\}$ is the outside demand for listing $i - 1$, and seat 5 never acts. Exactly $m = 2$ listings are lemons, $\Lambda \subset \{0, \ldots, 3\}$, drawn uniformly from a random stream separate from the one that sets values, asks and costs. Seat 0 signs at most one lease (ℓ, rent), or none. Its payoff after signing ℓ with k inspections is

$$u_A = v_\ell - D \cdot \mathbb{1}[\ell \in \Lambda] - \text{rent} - c \cdot k, \text{ and } u_A = -c \cdot k \text{ if it signs nothing,} \tag{A.1}$$

where D is the lemon loss and c the inspection fee (Table A.1).

Seat 0 privately knows v_ℓ, its value for each listing if the listing is sound. Each landlord privately knows its listing's quality and cost. The prior is uniform over the six possible lemon pairs. The landlord kind is not drawn: both kinds are played on every world.

Every landlord has a target rent: it accepts any offer at or above the target and answers a lower offer with a counter at the target (Appendix B.1). The two kinds differ only in where the target comes from. A *leaky* landlord prices off its own cost. A lemon costs its landlord $1,000 less, so a lemon's landlord will rent for about

$1,000 less than a sound listing's landlord, and an offer between the two rents is accepted only by a lemon's landlord: the acceptance itself shows that the listing is a lemon. A *pooled* landlord prices every listing as if it were sound, so its replies carry no information about quality. Each world is played once under each kind.

Table A.1. Housing parameters.

Symbol	Meaning	Value
T	rounds	3
m	lemons	2 of 4
D	lemon loss (cuts the tenant's value and the landlord's cost alike)	$1,000
c	inspection fee	$25
a_ℓ	ask	$\lfloor U(1500, 3200) \rfloor$
v_ℓ	seat 0's value if sound	$a_\ell + 10 \cdot s_\ell + U(-20, 20)$, with $s_\ell = 100(0.6\kappa_\ell + 0.4\iota_\ell) - 30$ and κ, $\iota \sim U(0, 1)$; κ_ℓ and ι_ℓ are listing attributes shown on the board, which move seat 0's value but say nothing about quality
$c^S{}_\ell$	a sound listing's cost	$\max(0, a_\ell - U(20, 80))$
$c^L{}_\ell$	a lemon's cost	$\max(0, c^S{}_\ell - 1000)$
μ	landlord margin: a landlord's target is its reservation plus μ, capped at the ask (B.1)	$25
β	outside-demand rate per round	½ (v15's holdout: 0.3)

In the main evaluation set, asks range from $1,501 to $3,198, and seat 0's gain $v - a$ ranges from −$275 to +$676.

Protocol. Each round $r = 0, 1, 2$ has four phases. (1) *Inspect*: seat 0 may pay c to inspect one open listing it has not yet inspected, and the quality appears in its observation at once. (2) *Contact*: seat 0 makes one offer (ℓ, x) with ℓ open and $x \geq 0$, or passes; offering again on a listing it walked away from is legal. (3) *Respond*: each landlord applies its policy (Appendix B.1) to its inbox, and each listing creates at most one binding hold. (4) *Commit*: a seat holding a hold signs or walks; a missing or invalid action counts as a walk, and every hold expires at the end of the phase. Signing ends seat 0's play, and the market ends after round 2 or when every listing is leased.

Observation of seat 0. Seat 0 observes the board (asks, uninformative listing attributes, and whether each listing is open or leased), its rejected listings and active hold, its values v, the lemon count and share, the loss, the fee, its own inspections and spending, and a history field (`your_history`) that lists its own offers, each hold with the binding rent and whether the landlord accepted or countered, and each sign or walk. The sealed prompt states the payoff, that a fixed landlord "may accept or make one binding counteroffer", that "a counteroffer can itself convey information about listing quality", the departure rule, that a listing seat 0 bid on is protected in that round, the three-round horizon, that walking does not close a listing, and the order of the phases. The v14 prompt states neither the landlord's pricing rule nor the cost range. The v15 task prompt states both: a sound landlord accepts any offer at or above its cost plus $25 (or the ask, if lower) and otherwise counters at that amount, and since a sound listing's cost is $20 to $80 below its ask, a rent more than $55 below the ask proves a lemon.

Example: a reply and a signing. A listing asks $a = $2,400. Its sound cost is $c^S = $2,350, so a lemon's cost is $c^L = $1,350, and seat 0 values the listing at $v = $2,600 if sound. With the $25 margin (Appendix B.1), a sound listing's landlord targets $2,375, a lemon's leaky landlord $1,375, and a lemon's pooled landlord $2,375. Suppose seat 0 offers $2,000. A sound landlord counters at $2,375, and a pooled lemon landlord also counters at $2,375; a leaky lemon landlord accepts at $2,000, and only a lemon's landlord would, so the acceptance it-

self identifies the lemon. Signing pays v − rent = +$225 for the sound listing at $2,375, v − D − rent = −$400 for the leaky lemon at $2,000, and −$775 for the pooled lemon at $2,375. Inspecting first costs $25 and reveals the quality under either kind. Had seat 0 offered $0, the leaky lemon landlord would have countered at $1,375, which also identifies the lemon, and signing at that rent pays +$225.

A.2 Procurement: pay or test (the earlier evaluation set)

Environment. One buyer sources one component. The suppliers are one recorded supplier E (an established supplier with a sales record) and three newcomers N1–N3. Each supplier has a type $\theta \in \{G, L, P\} = \{$(on-time 0.97, yield 0.97), (0.55, 0.97), (0.97, 0.55)$\}$, that is, good, late or poor-yield; E is always of type G. Newcomer types are independent. Newcomer i's prior is read from the cohort record of its seller kind, in which a_i of 40 tracked sellers were unreliable on delivery and b_i failed inspection:

$$\pi_i = (\gamma_i, \lambda_i, \rho_i) = ((40 - a_i - b_i)/40,\ a_i/40,\ b_i/40). \tag{A.2}$$

The three newcomers therefore have $3^3 = 27$ type profiles, each with positive probability, and γ ranges from 0.2 to 0.7.

Table A.2. Procurement parameters.

Quantity	Value
Kits owed / minimum service	50 / 40 (an award completing fewer than 40 is void)
Revenue per kit / shortfall penalty	$37 / $12 per kit short
Supplying nothing, D	−$600 (= −12·50), plus any information already spent
Cash budget	$2,500 (never binds)
Deadline T	10–12 days (clock "none"), 13–14 (tight), 17–18 (loose)
Action budget	10
Order sizes	50, 55 or 60 units (minimum order quantity, MOQ, 50; step 5; capacity 60); shipping $0.60 a unit
E	price U[24, 28] rounded to $0.05; lead 6 days; sample $30 and 2 days; a record of 48–96 orders, 3% late, 3% defective
Newcomer i	price $p_E(1 - \delta_i)$, δ drawn without replacement from {3, 5, 7, 9, 12, 15, 18, 21}%; lead 6–8 days; sample $30–80 in $10 steps and 3 days
Cohort record	$a_i \in \{4, 8\}$, $b_i \in \{8, 14, 20, 24\}$ of 40

Payoff. An award of q units at price p from a supplier with yield y, on-time rate r and lead time ℓ, made on day t after spending I on information, is worth

$$M = 37{\cdot}k - 12{\cdot}(50 - k) - C \text{ if } k \geq 40 \text{ and the order is valid, and } M = D - I \text{ otherwise,} \tag{A.3}$$

where the number of kits completed is

$$k = \min(\lfloor q{\cdot}y{\cdot}r \rfloor, 50) \text{ if the order arrives by the deadline } (t + \ell \leq T), \text{ and } k = 0 \text{ if it arrives late,} \tag{A.4}$$

and the cost C is the sum of the terms in the following table:

Term of C	What it is
q·p	the purchase
+ 0.6·q	shipping, $0.60 a unit
+ q·p·0.12·15/365	financing the purchase at 12% a year for 15 days: the 45-day working-capital horizon less the 30-day payment terms

Term of C	What it is
− R	the expected refund on defective units
+ R·0.12·10/365	financing the refund, which arrives 10 days after the claim
+ I	the information bought

Here R is the expected refund on defective units: of the q·(1 − y) defective units, 90% of claims are accepted and refunded at the unit price, so

$$R = 0.9 \cdot q \cdot (1 - y) \cdot p. \tag{A.5}$$

For a good supplier, an order of q = 55 units is optimal at every price in the evaluation set, and before information costs the award is worth

$$A(p) = 1817 - 53.79p. \tag{A.6}$$

With 55 units at y = r = 0.97 all 50 kits are completed, so 1817 = 37·50 − 0.6·55 and 53.79 = 55(1 + 0.12·15/365) − 1.485(1 − 0.12·10/365), where 1.485p = 0.9·55·0.03·p is the expected recovery on defective units.

Protocol. The buyer takes one action at a time until it takes a terminal action (an award or a deferral, Table A.3), exhausts the action budget, or takes an invalid action. A deferral, an exhausted budget and an invalid action are each worth D − I.

Table A.3. Procurement actions.

Action	Cost	Reveals
request a quote	$2, 1 day	a binding offer at the listed terms and the true on-time rate, which identifies L
request a sample	the listed fee and days	100 inspected units, yield 0.97 or 0.55, which identifies P; no noise
inquire	$1, 1 day	verbal claims; every seller claims 0.97 and 0.97
counter-offer	$1, 1 day	nothing in this evaluation set (B.2)
check an award	$0, 0 days, one action	a dry run of the award; nothing beyond what quotes and samples showed
submit an award	terminal	eligible only with an unexpired formal offer, a verified sample and the exact variant (the component version the order requires)
defer	terminal	worth D − I

Every listing shows a claimed yield of 0.97 regardless of the supplier's type. The two task prompts differ in two lines: the second adds that the buyer "places many orders like this one and is judged on its average margin across them", and that "a formal quote is the supplier's final price and terms. A counter_offer cannot improve them."

Example: what a quote is worth. The recorded supplier lists $26.00, so awarding it 55 units earns A(26.00) = 1817 − 53.79·26.00 = $418.46. A newcomer lists 15% less, $22.10, which is worth A = $628.24 if the newcomer is good. Its cohort record shows 8 of 40 sellers late and 14 of 40 failing inspection, so γ = 0.45, λ = 0.20 and ρ = 0.35, and its sample costs $50. A quote costs $2 and identifies a late supplier, which saves the $50 sample with probability λ, so the quote is worth 0.20·50 − 2 = $8 before the sample is bought. Testing the newcomer (a quote, then a sample if the quote is clean) instead of paying the premium is worth γ(628.24 −

418.46 + 30) − 2 − (1 − λ)·50 = $65.90 in expectation, where the $30 is the recorded supplier's sample that the buyer no longer needs, provided that a failed test still leaves time to return to the recorded supplier before the deadline.

A.2.1 The evaluation set reported in Section 5.1

Section 5.1 reports results on a harder evaluation set; the set described above (Table E.14) is its predecessor. The payoff, the three types, the kits owed, the action budget and the order sizes are unchanged. Table A.6 lists what changes.

Table A.6. What the harder Procurement evaluation set changes.

Quantity	Earlier set (above)	Harder set
Newcomer price	$p_E(1 - \delta)$, δ from 3% to 21%	$p_E(1 - \delta)$, δ drawn from the price gaps in our own sourcing record (−189% to +66%; a newcomer is dearer about half the time)
Shared source	none: newcomer types independent	newcomers of one seller kind share a source with probability $s \in \{0.5, 0.65, 0.8, 0.95\}$; if they do, all take one type drawn from their cohort's odds
Quote	always answered	a newcomer answers each request with probability 1/3, independently; an unanswered request costs its $2 and day and shows nothing; the recorded supplier always answers
Sample	$30 to $80, 3 days	$20 to $240, 3 days; always arrives
Source lookup	none	$1, 0 days: names the listing's registered factory; two newcomers of one kind name the same factory if and only if they share a source
Inquiry, counter-offer	offered; neither can raise the buyer's payoff	removed
Financing, cash budget, shipping	12% a year; $2,500; $0.60 a unit	none; never binds and is not shown; folded into the listed price
Cohort record	$a_i \in \{4, 8\}$, $b_i \in \{8, 14, 20, 24\}$ of 40	$a_i \in \{2, 4, 6\}$, $b_i \in \{18, 22, 26, 30\}$ of 40
Deadline	three clocks	14 to 22 days
Inclusion criterion	none	a rule that plans one newcomer's test exactly loses at least $10 to the best policy and pays the premium first; every fixed rule and a kind-blind shortcut lose at least $10; the best policy beats paying the premium by at least 10% of its value; without a shared source the one-step rule is within $5 of the best policy; information is worth at least $10

We compute the best informed policy exactly by backward induction over what the buyer has seen (each supplier's test results, which listings it looked up, and whether the sharing of each kind is settled), with an unanswered quote and a lookup treated as chance outcomes. Its value equals the environment's own average over 400 plays to within four standard errors in each of the 12 pilot worlds and each of the 20 worlds of the preregistered test. Those 20 worlds are the first included seed in each of 20 blocks starting at seed 9,400,000 (we scanned 5,164 seeds). Their twins at the earlier set's odds keep every listing, price, sample term, clock and sharing probability, and redraw only the cohort records.

A.3 Data center: negotiating into an agreement

Outcomes. An outcome is either an agreement, a clause vector c with an all-in price p, or no deal:

$$X = (C \times \mathbb{R}_{+}) \cup \{\varnothing\}. \tag{A.7}$$

Table A.4. The eight clauses.

Clause	Positions
warranty W	none, fix, fix and delay damages
damages rate d	50, 100, 200 a week (effective only under fix and delay damages)
liability cap K	uncapped, 1,500, 500
site readiness R	client, integrator
consequential loss Q	excluded, included
deposit δ	none, 25%, 50% of hardware
escrow E	no, yes (effective only with a deposit)
burn-in B	no, yes

After positions without effect are normalised away, $|C| = 5{\cdot}3{\cdot}2{\cdot}2{\cdot}5{\cdot}2 = 600$ distinct agreements remain, and every one of them can be reached by redlining. The negotiation starts from the integrator's standard draft c^0, which its playbook (coordination, managed or turnkey) sets; the integrator's private type in turn fixes the playbook (below).

Private types. The integrator's type $\theta_I = (t, \rho_I)$ consists of a pre-staging cost $t \in \{40, 120, 200\}$ and a risk charge $\rho_I \in \{0.3, 1.2\}$, and the six types are equally likely. The type fixes the playbook: $\rho_I \geq 1$ gives coordination; otherwise $t \leq 40$ gives turnkey, and every other type gives managed. The draft therefore narrows the client's prior, and in 22 of the 58 worlds of the development set (Appendix E.2) it reveals the type. The client has a risk charge $\rho_C \in \{0.15, 0.75, 1.55\}$, with a stated prior of one third on each value. In the evaluation sets, ρ_C is fixed by the world's stratum (one of eleven outside-option situations, Appendix E.2) rather than drawn.

Table A.5. Data-center symbols. Money is in $ thousands. Ranges are the generator's uniform draws in the base evaluation set (some strata widen them), and every quantity is public to both sides unless marked private.

Symbol	Kind	Meaning	Value or range
z_{pre}, z_{site}	decision (0/1)	the integrator pre-stages the cluster; it prepares the client's site (only if it carries site readiness)	chosen to minimise its cost
e_{def}, e_{late}, e_{inc}, e_{fail}	event indicator (0/1)	a compatibility defect; a late facility; an incident after handover; the integrator fails and the deposit is lost	independent given z_{pre}, z_{site} and the clauses
q_0, q_1	probability	a defect without and with pre-staging	0.25–0.35; 0.04–0.08
u	probability	a late facility (×0.4 if the site is prepared)	0.20–0.30
ι	probability	an incident (×0.5 with burn-in)	0.04–0.06
ν	probability	the integrator fails (counts only with a deposit not in escrow)	0–0.015
F	money	fixing a defect	200–300
S	money	standby cost of a late facility	150–220
Φ	money	an incident's loss	2,500–3,500
v	money per week	the client's cost of a week's delay	90–130
w_{def}, w_{late}	duration (weeks)	delay from a defect; from a late facility	3–5; 2–4
H, Ψ	money	hardware; services	9,600; 600

Symbol	Kind	Meaning	Value or range
δ	share	deposit, as a share of hardware	0, 0.25 or 0.5 (clause)
t	money, private	the integrator's pre-staging cost	40, 120 or 200
ρ_I, ρ_C	risk charge, private	each side's markup on the expected loss it carries	0.3 or 1.2; 0.15, 0.75 or 1.55
x	money	preparing the client's site	30–90
r_C	rate per year	the client's cost of capital on a deposit (13 weeks)	0.05–0.07
team	money	the client's own team if it manages the deployment	600–1,000
T	money	the rival turnkey offer	stated in both task prompts
O_I	money	the integrator's outside option	500
ω, Y	money, per outcome	what the contract says the integrator owes; what it pays after the cap K	below

Events and costs. Given z_{pre} and z_{site}, the four events are independent, with probabilities

$$P(e_{def}) = q_1 \text{ if the integrator pre-stages and } q_0 \text{ otherwise; } P(e_{late}) = u\cdot(0.4 \text{ if the site is prepared, else } 1); \; P(e_{inc}) = \iota\cdot(0.5 \text{ with burn-in, else } 1); \; P(e_{fail}) = \nu \text{ with a deposit and no escrow, and } 0 \text{ otherwise.} \tag{A.8}$$

In each of the 16 outcomes the real loss is

$$L = e_{def}\cdot(F + w_{def}\cdot v) + e_{late}\cdot(S + w_{late}\cdot v) + e_{inc}\cdot\Phi + e_{fail}\cdot\delta H, \tag{A.9}$$

the integrator owes

$$\omega = e_{def}\cdot[F\cdot\mathbb{1}(W \neq \text{none}) + d\cdot w_{def}\cdot\mathbb{1}(W = \text{fix and delay})] + e_{late}\cdot S\cdot\mathbb{1}(R = \text{integrator}) + e_{inc}\cdot\Phi\cdot\mathbb{1}(Q = \text{included}), \tag{A.10}$$

and it pays

$$Y = \min(\omega, K), \tag{A.11}$$

the cap applying to the total owed in each outcome. With $\mathbb{E}$ denoting the expectation over the outcomes, the integrator's cost C_I is the sum of the terms in the following table:

Term of C_I	What it is
$H + \Psi$	hardware and services
$+ z_{pre}\cdot t + z_{site}\cdot x$	pre-staging, if chosen; preparing the site, if chosen and it carries readiness
$+ 40\cdot B$	the burn-in crew, if the contract has burn-in
$+ (1 + \rho_I)\cdot\mathbb{E}[Y]$	what it expects to pay under the contract, marked up by its risk charge
$+ \mathbb{1}(R = \text{integrator})\cdot 1.5\cdot P(e_{late})\cdot S$	a surcharge on expected standby it carries but cannot control
$- \mathbb{1}(\text{deposit, no escrow})\cdot 0.14\cdot\delta H\cdot 13/52$	the return on a deposit it holds for 13 weeks at 14% a year

The integrator's best response (z_{pre*}, z_{site*}) minimises C_I, and we write its cost under that response as

$$IC(c) = C_I(z_{pre*}, z_{site*}). \tag{A.12}$$

The client's cost excluding the price, $C_C(c)$, is the sum of the terms in the next table:

Term of C_C	What it is
$(1 + \rho_C)\cdot\mathbb{E}[L - Y \text{ given } z_{pre^*}, z_{site^*}]$	the loss the contract leaves with the client, marked up by its risk charge
$+ \mathbb{1}(\text{deposit})\cdot r_C\cdot\delta H\cdot 13/52$	the cost of capital on its deposit
$+ 10\cdot\mathbb{1}(\text{escrow})$	the escrow fee
$+ v\cdot\mathbb{1}(\text{burn-in})$	a week's delay for burn-in

In these tables, B is 1 when the contract includes burn-in (Table A.4), whose crew costs the integrator 40; 1.5 is the integrator's surcharge on expected standby it cannot control, paid when it carries site readiness; 0.14 is its capital rate per year on a deposit it holds without escrow; 13/52 is the thirteen weeks between signing and delivery, as a share of a year; and 10 is the client's escrow fee.

Outside options and payoffs. The client's outside option is the cheaper of a rival turnkey offer T and managing the deployment itself:

$$O_C = \min(T, M), \text{ with } M = H + \text{team} + (1 + \rho_C)[q_0(F + w_{def}\cdot v) + u(S + w_{late}\cdot v) + \iota\Phi]. \tag{A.13}$$

The integrator's outside option is $O_I = 500$. Let n_i be the number of redlines side i makes after the first move. A signed agreement (c, p) pays

$$U_I = p - IC - O_I - 25\cdot n_I \text{ and } U_C = O_C - p - C_C - 25\cdot n_C, \tag{A.14}$$

and no deal pays each side $U_i = -25\cdot n_i$.

Protocol. The negotiation has at most six moves, and the integrator makes the odd-numbered moves. On move 1 the integrator redlines or walks, since no price is yet on the table. On moves 2–5 the side to move redlines, signs the other side's standing version at its price, or walks. On move 6 the client signs or walks. A redline sets any clauses to listed positions, states a positive all-in price, and may carry a note of up to 400 characters. A move that fails to parse or is illegal ends the episode as an invalid action; each move gets one attempt. Each side sees its own task prompt (every public number, its own type, and the other side's possible types with their odds) and, at each move, the document with the latest tracked change on each clause, the price on the table, and the full history including notes.

Example: one clause. Take $v = 110$, $\iota = 0.05$ and $\Phi = 3{,}000$ with consequential loss excluded, so the client carries incidents. Adding burn-in costs the integrator its crew, 40, and the client a week's delay, $v = 110$. Burn-in halves the incident probability, cutting the client's expected loss by $0.025\cdot 3{,}000 = 75$ before its risk charge, which the client values at $(1 + \rho_C)\cdot 75$: 86 at $\rho_C = 0.15$, 131 at 0.75 and 191 at 1.55. Joint cost therefore changes by $40 + 110 - (1 + \rho_C)\cdot 75$, that is, by $+64$, $+19$ or -41. Burn-in belongs in the best agreement only for a client with the high charge. The integrator does not know the client's charge, which is why the best agreement depends on the other side's type.

Appendix B. Counterpart policies

B.1 Housing: landlords and outside demand

Landlord. Each landlord has a reservation rent ρ_ℓ: $\rho_\ell = c_\ell$ (its own cost) under the *leaky* kind, and $\rho_\ell = c^S_\ell$ for every listing under the *pooled* kind. Its target rent is

$$\tau_\ell = \max(\rho_\ell, \min(a_\ell, \rho_\ell + \mu)). \tag{B.1}$$

Given the highest offer x* in its inbox (ties go to the lower seat), the landlord accepts, creating a hold at x*, if $x^* \geq \tau_\ell$, and otherwise counters at τ_ℓ. A \$0 offer therefore always draws a counter at τ_ℓ. In the main evaluation set, a sound listing's target lies in [a − 55, a], a lemon's target under the leaky kind lies in [a − 1055, a − 996], and a lemon's target under the pooled kind equals that of a sound listing. A reply reveals quality only to an offer below the sound floor $\min(a_\ell, c^S_\ell + 25)$; any offer at or above the floor is accepted whether the listing is sound or a lemon.

Outside demand. Listing ℓ is wanted in round r if and only if

$$H(\text{w-asks}, \ell, r)/2^{64} < \beta, \tag{B.2}$$

where H is the first eight bytes of a SHA-256 hash of the world's asks, the listing and the round, read as an integer. A wanted open listing is taken: its outside seat offers \$0, the landlord counters at τ_ℓ, and the seat signs in the same round. If seat 0 made an offer on ℓ in that round, its offer wins the tie and ℓ is not taken. The schedule depends neither on quality, nor on the landlord kind, nor on seat 0's play; in the main evaluation set, the listing is taken in 52.1% of listing-rounds.

B.2 Procurement: suppliers

Quotes carry the true base terms, which equal the listing, and samples are exact. A counter-offer is accepted only if every term lies between the supplier's limit and the current offer. Because every limit equals the quoted value (price floor = quoted price, minimum order 50, 30-day payment and refund windows), only a restatement of the current terms is accepted, so quotes are final. Awards are scored on the true private terms.

B.3 Data center: the scripted negotiator

The script plays seat j (the other side is −j). It knows its own type exactly, and it knows the other side only through the stated types Θ and their prior β_0. The shares s and s_{min} are fractions of the value on the table, Π; τ, χ and V are amounts of money; and r, α and κ are dimensionless weights. For a type θ and an agreement c at price p, we write g_j and g_{-j} for the two sides' surpluses over their outside options, and $G(c; \theta) = g_j + g_{-j}$ for their sum, which does not depend on the price. On each of its moves the script takes the five steps below.

Step 1: belief. The belief β is the prior β_0 restricted to the types for which every redline the other side has made was worth at least that type's outside option, renormalised. An update that would leave no type is skipped.

Step 2: value on the table. The value on the table is the largest expected joint surplus over agreements, floored at zero:

$$\Pi = \max(0, \max_c E_\beta G). \tag{B.3}$$

Step 3: ask. The script makes N proposals of its own (3 as integrator, 2 as client). With $\Delta = (s_{open} - s_{final})/(N - 1)$, its ask share starts at $s_1 = s_{open}$ and then moves according to

$$s_{k+1} = \max(s_{final}, s_k - \Delta\cdot\kappa_k), \text{ with } \kappa_k = (1 - r) + r\cdot\min(2, \max(0, \eta/\Delta)), \tag{B.4}$$

where η is how far the other side has moved toward the script since the script's last proposal, measured as a share of Π. The collapse style holds s_{open} until its last proposal and then asks s_{final}.

Step 4: proposal. For each set consisting of the most valuable remaining types, the script picks the clause set that maximises expected joint value over that set (the price-only style keeps the standing clauses). It prices that clause set at its own break-even plus the ask share of that value, caps the price at what the least willing type in

the set would accept, and rounds it to $5k. It then keeps the type set whose proposal has the largest expected value to itself, counting types outside the set as refusing. V is that largest expected value: the script's surplus from its best proposal, with a refusal counted as 0.

Step 5: move. Let τ be the expected value to the script of the standing offer, and let the value of proposing be

$$\chi = \alpha \cdot V - f, \tag{B.5}$$

where α is the probability the script assigns to its proposal being signed and f is the redline fee (25, or 0 on move 1). The script signs if $\tau \geq \max(0, s_{min} \cdot \Pi)$ and either it cannot propose or $\tau \geq \chi$. Otherwise it redlines if it has a proposal with $\chi > 0$ (a price-only script keeps redlining while it has a proposal), and otherwise it walks. It never reads or writes a note.

Table B.1. Counterpart styles of the data-center case. (Defaults $r = 1$, $\alpha = 0.8$, $s_{min} = 0$.)

Style	Role	Opening ask s_{open}	Final ask s_{final}	Least share s_{min}	Reciprocity r	α	Other
firm population (`drawn`)	core	U(0.60, 0.90)	max(0.45, s_{open} − U(0.05, 0.30))	U(0.10, 0.35)	0.5 or 1.0, equal odds	0.8	drawn per world and seat
soft	core	0.55	0.25	0	0	0.4	concedes a fixed amount each round regardless of the opponent's moves
price only	adversarial	0.70	0.50	0.20	1	0.8	never changes a clause
collapse	adversarial	0.90	0.30	0	unused	0.8	holds its ask, then concedes in its final proposal
overreach	adversarial	1.30	0.70	0.20	0	0.8	priced beyond what any type of the other side accepts while $s \geq 1$
prober	adversarial (information)	as the firm population					asks in notes for costs and a confidential item; accepts a stated cost that matches a type

1. **Firm population.** Each world and seat draws its own style, so the counterpart is a population of policies rather than one rule. In about half of the world-seat pairs ($r = 1$) it concedes no faster than the model does, and it refuses any offer below its least share. This style tests whether further negotiation increases the model's payoff against a counterpart that holds firm. In the development runs the style was drawn from a hash of the whole world, including the client's hidden charge, so two worlds that differed only in that charge could face different counterparts (the model never sees the style). Both confirmatory runs draw the style from the public parts of the world only, and the signalling test used one fixed style (the firm style) so that the opening could not depend on the client's type.
2. **Soft.** Concedes a fixed amount each round regardless of the model's moves. It tests whether the model captures the concessions that a weak counterpart offers.

3. **Price only.** Never changes a clause and negotiates only the price. It tests whether the model improves the clauses itself.
4. **Collapse.** Holds a 90% ask until its last proposal and then asks 30%. It tests whether the model holds out for the final concession instead of accepting the high early ask.
5. **Overreach.** Opens with offers that are worse than walking away for every type of the model's side (for two proposals as integrator, for one as client), and then lowers its demands. It tests whether the model refuses an agreement with a negative payoff.
6. **Prober.** The firm population plus a fixed sequence of notes asking for the other side's costs and for a confidential third party. It reads the model's notes for a stated cost and, if the statement matches a stated type, restricts its belief to that type without verifying the statement. It tests what the model discloses when asked.

Two *reading* counterparts were used only in the pre-registered test. Each is the firm script with its belief over the client's charge reweighted by the client's first counter-offer, either through the price or through the number of protective clauses the client asks for. The price model is

$$p - \bar{\mu} \sim N(b_0 + b_1(\lim(\rho) - \bar{\mu}), sd^2), \tag{B.6}$$

where $\lim(\rho)$ is the client's limit at charge ρ and $\bar{\mu}$ is that limit averaged over the stated prior. The clause-count model is

$$\text{number of protective clauses asked for} \sim N(m_\rho, sd^2), \tag{B.7}$$

where m_ρ is the mean count at charge ρ. The parameters were fitted per model on earlier runs and frozen in the code.

Appendix C. Reference policies and baselines

The references are of three kinds, and a shortfall means something different against each. A *reachable policy* (Housing's `inspect_lowball`) uses only what the deciding seat may know; it shows a value that is achievable with that information, but it is not an optimum. The *best policy given the decider's information* (Procurement) is the exact optimum over what the buyer can learn, so a shortfall against it is the whole cost of the subject's decisions. A *full-information upper bound* (Housing's ceiling, the data-center best agreement) uses hidden state that no seat observes, so a shortfall against it also contains the value of information that no seat had.

C.1 Housing

Reachable reference (`inspect_lowball`). In each round, `inspect_lowball` acts as follows. If no open listing has been verified sound, it inspects the uninspected open listing with the largest $v - a > 0$. It then offers \$0 on an open listing it has inspected, choosing a sound one if it has one and otherwise a lemon, and among those the one with the largest $v - a$. It signs if and only if the listing was inspected and its known value (v, or $v - D$ for a lemon) covers the rent. It never signs a listing it has not inspected, and it does not weigh the inspection fee. It earns \$291.67 a world on the main evaluation set.

Ceiling. The full-information ceiling assumes known quality and no inspection fee, and it is reachable in round 0:

$$C(w, \text{kind}) = \max(0, \max_\ell [v_\ell - D \cdot \mathbb{1}[\ell \in \Lambda] - \tau_\ell^{\text{kind}}]), \tag{C.1}$$

where τ_ℓ^{kind} is the landlord's target under that kind (Equation (B.1)). It averages \$447.82 under the leaky kind and \$292.40 under the pooled kind, with a mean of \$370.11.

Scripted baselines. `inspect_then_sign` inspects a listing when $(1 - p_d)(v - a) > c$, where $p_d = (2 - \text{lemons found})/(4 - \text{listings inspected})$ is the stated probability that a listing not yet inspected is a lemon (Equation (D.1)); it then offers $\min(v, a + 1)$ on the best verified-sound listing and signs it, earning \$144.70 a world. `sign_anything` never inspects, offers the ask on its favourite listing if that listing is sound, and signs any hold that its value if sound covers. It earns −\$79.53 a world under either decision-value measure, but −\$348.76 in realised net. `pass` never acts and earns \$0. The decision-value measure is the one used for the control comparisons in Section 5 and Appendix E.4.

Best informed tenant (development). This tenant computes an exact optimal plan under the market's rules. Its belief covers the landlord kind and the lemon pair, and it is updated by inspections and, for the `pricing` tenant, by the replies the tenant hears. We compute the tenant's plan by backward induction over the three rounds, using five functions that follow the phases of one round:

- V, at the start of a round, chooses whether to inspect, and which uncertain listing;
- U, the contact step, chooses which listing to make an offer on, or none;
- W_ℓ averages over the replies the landlord of ℓ may give, when none has been heard yet;
- C_ℓ, the commit step, chooses whether to sign the hold on ℓ or walk;
- E values moving to the next round, while each open listing the tenant did not bid on may be taken by outside demand.

Table C.1. Symbols of the best informed tenant.

Symbol	Kind	Meaning
r	round	0, 1 or 2
O	set	the open listings
$q \in \{?, S, L\}^4$	record	each listing not inspected, inspected sound, or inspected lemon
$z \in \{–, N, L\}^4$	record	each listing's reply: none heard, a counter within \$55 of the ask, or a reply more than \$55 below the ask
$h = (\theta, \Lambda)$	hypothesis	the landlord kind and the lemon pair; the belief is uniform over those consistent with q and z, and $z_\ell = L \Leftrightarrow (\theta = \text{leaky} \wedge \ell \in \Lambda)$
λ_ℓ	probability	$P(\ell \in \Lambda)$ under the belief
ψ_ℓ	probability	$P(\theta = \text{leaky} \wedge \ell \in \Lambda)$, the chance that ℓ's reply will be low
R_ℓ	money	the rent the tenant expects on ℓ: the rent heard once heard; before that $a_\ell - 25$ for N and $\min(a_\ell, \max(0, a_\ell - 1050) + 25)$ for L. A sound landlord's cost is on average \$50 under the ask and a lemon's \$1,000 lower still, and +25 is the margin μ, so these are the average lowest rents
B, G	sets	listings protected this round (bid on); listings taken by outside demand

The `pricing` tenant puts probability ½ on each landlord kind; the `stated` tenant assumes the pooled kind, so for it the replies carry no information. Both always offer \$0. The five functions are defined as follows:

$$V(r, O, q, z) = 0 \text{ if } r \geq 3 \text{ or } O = \varnothing; \tag{C.2}$$

$$\text{otherwise } V(r, O, q, z) = \max\{\, U(r, O, q, z), \max \text{ over } \ell \in O \text{ with } q_\ell = ? \text{ and } 0 < \lambda_\ell < 1 \text{ of } [\, -c + \lambda_\ell \cdot U(r, O, q[\ell \mapsto L], z) + (1 - \lambda_\ell) \cdot U(r, O, q[\ell \mapsto S], z)\,]\, \}; \tag{C.3}$$

$$U(r, O, q, z) = \max\{\, E(r, O, q, z, \varnothing), \max \text{ over } \ell \in O \text{ of } W_\ell(r, O, q, z)\, \}; \tag{C.4}$$

$$W_\ell(r, O, q, z) = C_\ell(r, O, q, z) \text{ if } z_\ell \neq –, \text{ otherwise } (1 - \psi_\ell) \cdot C_\ell(r, O, q, z[\ell \mapsto N]) + \psi_\ell \cdot C_\ell(r, O, q, z[\ell \mapsto L]); \tag{C.5}$$

$$C_\ell(r, O, q, z) = \max\{\, v_\ell - D\cdot\lambda_\ell - R_\ell,\ E(r, O, q, z, \{\ell\}) \,\}; \tag{C.6}$$

$$E(r, O, q, z, B) = 0 \text{ if } r = 2, \text{ otherwise } \Sigma \text{ over } G \subseteq O\setminus B \text{ of } 2^{-|O\setminus B|}\cdot V(r + 1, O\setminus G, q, z). \tag{C.7}$$

Before a landlord has answered, the tenant cannot know its listing's lowest rent, because a sound listing's cost lies anywhere from $20 to $80 under the ask. The tenant therefore plans with the average ($50 under the ask, plus the $25 margin) and replans with the true rent once an answer arrives. On the main evaluation set the `pricing` tenant earns $307.32 a world and the `stated` tenant $266.89, which is below the reachable reference, so `stated` is not a best policy for the v14 task prompt.

C.2 Procurement

State. The state is $s = (\sigma, n, t)$, where $\sigma \in \{U, Q, S, K, B\}^4$ records each supplier as untouched, quoted (on time), sampled (good yield), qualified or bad; n is the number of actions left; and t is the elapsed day, capped at $\bar{t} = T - \min \text{lead} + 1$. A quote moves a supplier from U to B with probability λ and to Q otherwise, and from S to B with probability $\lambda/(\gamma + \lambda)$ and to K otherwise. A sample moves a supplier from U to B with probability ρ and to S otherwise, and from Q to B with probability $\rho/(\gamma + \rho)$ and to K otherwise.

Recursion. The value function satisfies

$$V(\sigma, 0, t) = D, \tag{C.8}$$

$$V(\sigma, n, t) = \max\{\, D,\ \max_{i \text{ with } \sigma_i = K} A_i(t),\ \max_{(a,i) \in R(\sigma)} Q(\sigma, n, t; a, i) \,\}, \tag{C.9}$$

$$Q(\sigma, n, t; a, i) = -c(a, i) + \Sigma_{\sigma'} P(\sigma' \mid \sigma, a, i)\cdot V(\sigma', n - 1, t + d(a, i)), \tag{C.10}$$

where the last term of Equation (C.9) applies only when $n \geq 2$ and $t < \bar{t}$. Here $R(\sigma)$ is the set of quotes on suppliers in U or S and samples on suppliers in U or Q, and $A_i(t)$ is the best award margin with good terms (D if the supplier cannot deliver by T). V counts only spending still to come, because information already bought is sunk. Ties go to the award, then to a request, then to deferral. On all 24 worlds, $V(U^4, 10, 0)$ equals the environment's average over the 27 type profiles to within $3\cdot10^{-13}$, and its mean is $450.86.

Fixed rules. A rule's expected loss is the equal-weight mean of $V^* - V^{\text{rule}}$ over the 24 worlds.

Table C.2. Procurement fixed rules.

Rule	Definition	Expected loss per world
one step ahead	finish any newcomer test begun; otherwise quote the untouched newcomer with the largest gain $\gamma(A_i - A_E + c_E) - 2 - (1 - \lambda)f_i$ (f_i newcomer i's sample fee, c_E the recorded supplier's, which a qualified newcomer saves) if positive and if a failure would still leave the recorded supplier time to deliver; otherwise qualify and award E	0.06
test the best odds once	quote then sample the newcomer with the highest γ; award it if qualified, else go to E	45.72
always pay	quote, sample and award E only	75.19
test the cheapest once	as above, for the cheapest newcomer	81.38
work up the price list	newcomers in price order until one qualifies, E only when none is left	146.11

For this appendix we also compute the bound that knows each newcomer's type; it averages $513.39 over the 27 profiles.

C.3 Data center

Best agreement. The joint cost of an agreement and the best agreement are

$$J(c) = IC(c) + C_C(c), \text{ and } c_* = \text{argmin over the 600 agreements of } J(c), \tag{C.11}$$

with ties broken in favour of the draft and then by a fixed order. The value available is

$$A = \max(0, O_C - O_I - J(c_*)). \tag{C.12}$$

Pairs. In the *oracle pair*, the integrator proposes c* at IC* + O_I + A/2 (or walks if A ≤ 0) and the client signs; its joint value lost is 0 on all 58 worlds. In the *draft pair*, the integrator proposes c^0 unchanged at its midpoint price, which loses $318k of joint value a world over the 58 worlds.

Stand-ins against a script. Two stand-in policies know both true types and always redline to c* at a share h of A. The first, *takes the first offer*, signs the script's opening (as client) or its first counter (as integrator) if it is worth at least walking away; it earns $114k a world as client and $122k a world as integrator. The second, *best plan*, searches every sequence of {walk, sign, c* at h ∈ {0.1, …, 1.0}} against the deterministic script with its stated style and keeps the sequence that is best for itself net of fees. It is an exact best reply within that grid, so it serves as a ceiling rather than a target.

Best informed policy, turn by turn (computed after the runs). For either seat, this policy plays optimally given what that seat knows: its own costs, the stated odds of the other side's types, and the script's rule and style. We find it by backward induction over the turns and over the types still consistent with every script move so far. - **Its moves.** At each turn it walks, signs, or redlines to the agreement (or the standing terms) worth most in expectation to some set of types still consistent with the history, at a price on a grid over each type's bargaining range. At the integrator's last redline it also tries the highest price at which the client script still signs. - **Checks.** We replay every plan against the original script. - **A lower bound.** Because it searches a restricted class of plans, its value is a lower bound on the best attainable. Of 720 model plays across both confirmatory runs, 2 integrator plays and 7 client plays earn more than the same search does when it is told the other side's type. - **Loss by turn.** Each model move is charged the expected value it gives up relative to this policy, given what the model's side had seen. - **What it shows.** Knowing the other side's type is worth $0.1k to $2.7k a world to this policy, in either seat and in either confirmatory run.

Appendix D. Measures

Each measure answers one question in the case's own units.

D.1 Housing. *What did the tenant's decisions earn, with luck taken out?* The measure is the tenant's net in dollars, except that a listing signed without knowing its quality is valued at what the tenant could expect from it rather than at the quality that happened to be drawn. A decision is *blind* if the listing had not been inspected when seat 0 committed. The stated probability that such a listing is a lemon is

$$p_d = (2 - \text{lemons found})/(4 - \text{listings inspected}). \tag{D.1}$$

A blind signing is valued at

$$\text{value} = v - D \text{ if the landlord is leaky and the rent is below the listing's sound floor, and value} = v - D \cdot p_d \text{ otherwise.} \tag{D.2}$$

For each episode, the tenant's net is

$$N = v_{true} - \text{rent} - c \cdot k \text{ for an informed signing, and } N = \text{value} - \text{rent} - c \cdot k \text{ for a blind one,} \tag{D.3}$$

where v_{true} is $v - D$ for a lemon and v for a sound listing. For each world,

$$Y_w = \tfrac{1}{2}(N_{leaky} + N_{pooled}), \tag{D.4}$$

and a world missing either landlord kind is dropped. We report the mean of Y_w over the main evaluation set with a 95% Student-t confidence interval. Comparisons between models are paired by world, with p-values Holm-adjusted across the three pairs. We also report each model's difference from the reachable reference, and treat \$50 as the smallest meaningful difference. The measure has two known gaps: a counter at the sound floor proves a listing sound but is still valued at p_d, and a lemon revealed on one listing does not update p_d on another. The *corrected measure* values every blind signing at

$$v - D \cdot P(\ell \in \Lambda \mid \text{the inspections and every informative reply}). \tag{D.5}$$

In the v14 episodes it changes the value of 57 signings and no pre-registered verdict, and the second gap (a lemon revealed on one listing) has no effect on these data. Table 5.2 reports both measures.

D.2 Procurement. *Where in the episode was value lost, and how much?* Each action is charged what it gave up against the best action available at that point, given what the buyer had learned. The charges add up to the play's shortfall in expected value, so the measure says both how much was lost and at which step. For the action a_t taken in state s_t, the charge and a play's decision loss are

$$\ell_t = V(s_t) - Q(s_t, a_t), \text{ and } DL = \Sigma_t\, \ell_t. \tag{D.6}$$

For an inquiry, a counter-offer or an award check, $Q = -c + V(\sigma, n - 1, t + d)$. For a deferral or an invalid action, $Q = D$, the value of supplying nothing ($-\$600$, Table A.2). For an award, Q is the award's margin at $I = 0$, or D if the award is infeasible. Because every Q is an expectation, a policy's expected decision loss is its shortfall from the best policy:

$$E[DL(\pi)] = V^*(so) - V^{\wedge}\pi(so). \tag{D.7}$$

We report DL in five parts: requests; strays (inquiries, counter-offers and award checks); walking away (a deferral or an invalid action); award timing (V less the best award available); and award terms (the best award available less the award taken).

D.3 Data center. *How much of what the best agreement makes available did the pair fail to realise?* Value is lost by settling on worse clauses, by failing to agree, or by paying for redlines; how the remainder is split between the sides is reported separately, as each side's surplus. The joint value lost is

$$JVL = A - (U_I + U_C), \tag{D.8}$$

which is the sum of three parts: the allocation part, $J(c) - \min(J(c^*), O_C - O_I)$, if an agreement is signed; the no-deal part, A, if nothing is signed (an invalid move counts as no deal); and the delay part, $25 \cdot (n_I + n_C)$. Each side's surplus is U_I or U_C. A side signs below its walk-away point if its surplus before redline fees (its U plus 25 times its own redlines) is negative.

Code and data availability

All code, evaluation sets and sealed evidence are in the AERead repository (https://github.com/aeread-org/AERead). A replication note in the repository (`docs/research/paper_replication_note.md`) maps each formula in Appendices A–D, and each procedure the paper reports, to its implementation.

Appendix E. Case details

E.1 Every reported check

Table E.1. Every reported check in the four cases.

Case	Check	Kind	Output	Scope
Housing	tenant net per world against the reachable reference and the ceiling	value	dollars	outcome
Housing	holds signed that only a lemon's landlord would offer; ending on an uninspected listing	rule	count	path
Housing	model against model, paired by world	relative	dollars	outcome
Procurement	dollars given up against the best informed policy, action by action	value	dollars	path
Data center	joint value lost against the best of 600 agreements	value	dollars	outcome
Data center	each side's surplus, by counterpart	relative	dollars	outcome
Data center	signed below one's own walk-away point; disclosed a confidential item	rule	count	outcome; path
Refund	resolution (eligibility, remedy, amount, liability; workflow) against the reference	answer (implemented as rule checks in the code)	share of fields matched	outcome
Refund	evidence complete before deciding; contract format; payment bound to the proposal	rule	pass or fail per leaf	path

None of the four cases uses a rubric. Code computes every score from the recorded trajectory, with no model or human rater in the loop. The kernel's replay certifies the score recorded in each receipt. In Housing and Procurement, however, the headline measure is computed outside the receipt (Appendix F.2), so a contributed case should make its headline measure a receipt leaf that replay covers. The runtime supports rubric leaves, but it keeps them beside the computed leaves and never folds them into the computed score.

E.2 Construction of the evaluation sets

Each evaluation set is drawn from a generator and filtered by an inclusion criterion. Where a case has a hidden draw, the criterion reads only what the deciding side could observe, so inclusion cannot select on the draw. We name the strata before any model plays, and we draw held-out worlds from a seed range disjoint from the development set.

Housing. A seed is included when the tenant's favourite listing and a second listing would be worth more than their asking rents if sound. Each seed keeps its own lemon draw, and we read the stratum (the favourite is a lemon, or it is sound) only after inclusion. Seeds are taken in order until each quota is full: a main evaluation set of 200 favourite-lemon and 60 favourite-sound worlds, and a holdout of 30 and 10 worlds from a disjoint seed range. The quotas oversample favourite-lemon worlds, where the decision is hardest. This is v14's criterion. v15 (Section 5.1) applies it and then screens each candidate world offline: - **The screen.** The best informed tenant and `inspect_then_sign` play both landlord arms. A world is included only if its full-information ceiling is above $0 in both arms and the best informed tenant beats the rule by at least $25 on the corrected measure, averaged over the two arms. - **What it included.** The main evaluation set of 260 worlds (200 and 60) came from 437 seeds: 119 failed the screen, 6 had a $0 ceiling and 7 failed the structural criterion. - **The other**

streams. The holdout of 40 worlds runs with outside demand at 0.3 a round and comes from a disjoint seed range; a development set of 20 worlds serves the model-profile check and the variance pilot. - **Records.** No two included worlds are near duplicates, and the evaluation set's scan logs every seed's numbers.

Procurement. The earlier evaluation set contains six kinds of world, four worlds of each. In two kinds paying the premium is right, either on price alone or because a failed test would leave no time to buy from the recorded supplier. In the other four a test is right, either of the cheapest newcomer or of another newcomer, on a tight or a loose clock. We classify a world from its listings with every newcomer assumed good, and draw the newcomers' types afterwards from a separate random stream. The kinds are sampled evenly; if they were drawn as they fall, a test of the cheapest newcomer would be right in most worlds. This classification screens no shortcut, which is why a one-step rule nearly solves the evaluation set (Appendix F.4). The second evaluation set's inclusion criterion does screen shortcuts. Computed from what a buyer observes and the stated law, it keeps a world only if each screened shortcut, including the one-step rule and four fixed rules, gives up at least $10 against the best policy, and it requires the included worlds' mean one-step loss to reach $15; 24 worlds were included from 5,798 seeds. A sibling evaluation set of 18 worlds places a marketplace record on one component's listings, from which an agent can read which supplier to verify.

Data center. The development set has eleven strata, each with its own outside-option situation and its own set of allowed playbooks, and two worlds per stratum and playbook: 58 worlds in all, 52 with a deal worth signing and 6 in which no agreement beats both outside options. The 58 development worlds come from the generator of the case's predecessor. The pre-registered test used 58 fresh worlds from the same generator, none identical in content to a development world, and for its type-swap experiment it wrote 29 of them twice, identical except for the client's hidden risk charge. The inclusion criterion implemented for these runs is the predecessor's (a shortcut must lose by a margin on the playbook's menu); the agreement case's own criteria were not used. The confirmatory evaluation set is filtered by the agreement case's own criterion, computed from the public world and the pre-registered prior before any hidden type is drawn. In expectation over the prior, a world is included only if four conditions hold. The best agreement beats the draft by at least $75k. Knowing the client's hidden charge is worth at least $25k to the integrator's choice of agreement, and for every integrator type the best outcome changes with that charge. Signing the draft as it stands, moving only the price, and either side demanding every protection each lose at least $75k against the best agreement. Walking away loses at least $150k. We leave redline fees out of every shortcut's value, which favours the shortcut, so each margin is a lower bound. Of 2,729 seeds scanned, 60 were included (45 with a deal and 15 without); v1 used them, with types drawn after inclusion and a separate 12-world variance pilot (Appendix F.2). v2 (Table 5.4) kept these criteria and added one screen, computed before any episode: in a world with a deal, the integrator's full-information ceiling against the firm population must beat the price-only rule by at least $25k. Its seeds are disjoint from those of every earlier evaluation set; of 5,436 seeds scanned, 80 were included (72 with a deal and 8 without).

E.3 What was run

Table E.2. What was run.

Case	Worlds or cases	Strata	Subjects	Plays	Cost (list)	Status
Housing v14	260 main + 40 holdout	favourite lemon (200 + 30), favourite sound (60 + 10); both landlord kinds in every world	Gemini 3.8 Flash, GLM 5.3 Flash, GPT-6 Luna	1,800 episodes and a 48-episode fill	about $8	confirmatory

Case	Worlds or cases	Strata	Subjects	Plays	Cost (list)	Status
Housing v15	260 main + 40 holdout (outside demand 0.3), from a screened evaluation set	as v14	Gemini 3.8 Flash, GPT-6 Luna, and Luna's A/A twin	1,720 episodes, all verified by replay; pilot 160 episodes	$8.64 (pilot $1.08)	pre-registered; Gates 1, 2, 3 and 5 passed, 4 partial
Housing v15 addendum B	the v15 evaluation set: 260 main + 40 holdout	as v15	Sol 6.1 (Codex), Opus 5.5 (Claude Code), both isolated	1,200 episodes, all completed and verified by replay	$24.30 and $54.03 at list price (subscriptions)	pre-registered addendum, outside v15's contrasts
Housing, CLI agents	first 20 of the main evaluation set	10 and 10	Opus 5.5, Sol 6.1	40 episodes each	$4.72	development
Procurement, pay or test	24	six kinds × four	Opus 5.5, Sol 6.1	96 episodes over two task prompts	$9.10	development
Procurement, pay or test, flash models	24	six kinds × four	Gemini 3.8 Flash, GPT-6 Luna (GLM's route refused)	24 episodes each, second task prompt	$0.21	development
Procurement, record evaluation set	18	which listing the record favours	Opus 5.5, Sol 6.1	one episode a world each		development
Data center development and signalling	58 development + 58 fresh	deal or no deal; outside-option kind; counterpart style	Opus 5.5, Sol 6.1	2,456 negotiations, 2,417 valid	$119.30	development; one pre-registered test
Procurement v2 gate and pilot	12 pilot worlds, two plays per model; one gate each	shared-source information value	Gemini 3.8 Flash, GPT-6 Luna	50 episodes	$0.2193 charged	stopped at sizing; no confirmatory run
Data center confirmatory v2	80 included + 12 pilot	72 deal / 8 no-deal; both tools isolated	Opus 5.5, Sol 6.1	640 negotiations, all valid and replayed	$55.41	pre-registered; Gate 3 passed, the others partial
Data center confirmatory v1	60 included + 12 pilot	45 deal / 15 no-deal in the confirmatory run	Opus 5.5, Sol 6.1	1,176 model negotiations, 1,175 valid; 120 additional scripted dry runs	$74.41	pre-registered primary contrasts; eight departures
Refund V2.5	180 distinct cases (seed 301)	eleven cohorts × four seats	deterministic reference profiles only	816 trajectories	none	developmental release
Refund V2.1	6 cases	one policy seat	DeepSeek V4 Flash, Gemini 2.5 Flash Lite, GPT-5.6 Luna, Grok 4.3	480 episodes (20 repeats of each case)		descriptive

Opus 5.5 ran through Claude Code and Sol 6.1 through the Codex CLI, without tools, at low reasoning effort, with one subprocess per action. Neither CLI exposes a seed or a temperature, so we record both settings as unavailable rather than assume values for them. In the v1 confirmatory run the two CLIs were not equally isolated from the operator's machine (Table H.1); for v2 we isolated both and probed their context (Section 5.1). Refund V2.5 makes no paid model call. Its release qualifies the environment with deterministic profiles in

every seat and replays every trajectory, so it measures the environment's conformance rather than a model. With every seat scripted, its 816 trajectories reduce to 174 distinct action sequences (four payments cases share one reference trajectory).

E.4 Results beyond the main text

Housing

Section 5.1 and Table 5.2 report v15, the pre-registered comparison. Its inclusion screen rejected 119 of the 437 candidate main worlds. Relative to the best informed tenant's value, the full-information ceiling is 121% and `inspect_then_sign` earns 49%. The A/A arm, in which Luna runs twice under two names with different seeds, differs by −1.6 points [−6.1, +2.9], inside the $50 band, which indicates that the band is calibrated. In the holdout, Gemini leads Luna by 18.2 points [5.8, 30.6]. In worlds whose favourite is a lemon, Luna signs without inspecting twice as often as Gemini (230 against 114 blind signings in 600 episodes). All 1,720 episodes completed and were verified by replay ($8.64). The evidence bundle also records 0 invalid seat-0 actions for either model, and realized nets of $244.0 (Gemini) and $194.2 (Luna) beside the corrected-measure values; the v15 results note in the repository gives every row of Table 5.2 in dollars.

The addenda. GLM 5.3 Flash was excluded at the pre-run profile check, because its provider now serves a different quantization from the one v14 pinned. A pre-registered addendum ran it on another provider serving v14's quantization (600 of 600 episodes, verified by replay), and we report it beside v15 and outside its pre-registered contrasts. On the main evaluation set GLM keeps 53.9% of the best informed tenant's value. It does not beat `inspect_then_sign` at the pre-registered 98.3% level (+4.5 points [−4.4, +13.3]). Descriptively, it trails Luna by 8.9 points [1.6, 16.2] and Gemini by 24.8 [19.0, 30.5]. A second pre-registered addendum ran Sol 6.1 and Opus 5.5 through their command-line tools on the same 600 episodes each, all completed and verified by replay, likewise outside v15's contrasts. Sol keeps 90.4% of the best informed tenant's value and comes closest to `inspect_lowball` (5.0 points below it); Opus keeps 80.4%. Both beat `inspect_then_sign` at the 98.3% level. Sol leads Opus by 10.0 points [7.4, 12.6], and Opus cannot be distinguished from Gemini (+1.6 [−0.3, +3.6]). As with the flash models, nearly all of each model's shortfall arises in worlds whose favourite is a lemon.

The earlier comparison (v14). The rest of this subsection reports v14, the earlier comparison of three models (Table E.13). v14's pre-registered measure conditioned on only one landlord kind.

Table E.13. Housing v14, main evaluation set, in dollars per world with 95% intervals over worlds. The corrected block is a post-run reanalysis that we reproduced from sealed records on 16 October; the pre-registered measure remains visible.

	Gemini 3.8 Flash	GLM 5.3 Flash	GPT-6 Luna
Complete worlds: all (favourite lemon / sound)	260 (200 / 60)	260 (200 / 60)	260 (200 / 60)
Pre-registered measure, as frozen			
Tenant net	+211 [+189, +233]	+137 [+111, +162]	+130 [+104, +156]
Minus the reference, favourite is a lemon	−101 [−120, −82]	−179 [−203, −154]	−207 [−229, −186]
Minus the reference, favourite is sound	−14 [−18, −9]	−76 [−116, −36]	−9 [−16, −2]
Minus a rule that inspects when the odds make it pay, then signs	+66 [+50, +82]	−8 [−30, +14]	−15 [−32, +2]
Corrected measure, reanalysis			
Tenant net	+213 [+191, +235]	+149 [+124, +175]	+147 [+121, +174]

	Gemini 3.8 Flash	GLM 5.3 Flash	GPT-6 Luna
Minus the reference, favourite is a lemon	−98 [−117, −80]	−164 [−189, −139]	−185 [−208, −162]
Minus the reference, favourite is sound	−14 [−18, −9]	−70 [−108, −32]	−9 [−16, −2]
Minus a rule that inspects when the odds make it pay, then signs	+68 [+52, +84]	+5 [−17, +27]	+3 [−15, +20]

Under the analysis pre-registered before any episode, which compares three pairs with Holm adjustment and treats $50 as the smallest meaningful difference, Gemini leads GLM by +74 [+53, +95] dollars a world and Luna by +81 [+63, +98], and GLM and Luna do not differ (+7 [−16, +29]); the 40-world holdout agrees. Under the corrected measure, a reanalysis that was not pre-registered, the leads are +63 [+43, +84] and +65 [+49, +82], and GLM minus Luna is +2 [−20, +24]. No pre-registered verdict changes, but both lower bounds fall below $50. No model reaches the reference, and almost the entire shortfall arises in worlds where the favourite listing is a lemon. The rule in the last row of each block, which we scored after the runs, earns $145 a world; under either measure Gemini beats it, and GLM and Luna cannot be distinguished from it.

On all 260 worlds of the main evaluation set, with Gemini's refused episodes taken from the pre-registered fill, the models fall short of the reachable reference by −81 [−96, −66] (Gemini), −155 [−177, −134] (GLM) and −162 [−181, −142] (Luna) dollars a world. They keep 60%, 27% and 22% of what the reference earns where the favourite is a lemon, against 97%, 83% and 98% where it is sound. The 40-world holdout points the same way (Gemini−GLM +105 [+46, +164], Gemini−Luna +65 [+23, +106], GLM−Luna −40 [−106, +26]). The original 1,800 published receipt rows comprise 1,764 scored completions and 36 exclusions for operational failure. Replay reconstructs each completed receipt's score, which totals all six tenants; failure rows are verified as exclusions rather than rescored (Appendix H.3). The derived analyses re-derive the seat-0 measure of Table E.13 from the sealed event logs.

Scripted tenants, played offline on every episode, give the floor. Passing earns $0. Never inspecting, and signing any hold that the listing's value would cover if it were sound, earns −$80 a world, because this rule signs lemons. Inspecting when the stated odds make it pay and then signing earns $145. The models beat the first two rules by $130 to $292 a world. Sixteen GLM episodes contain invalid tenant actions, almost all of them offers on a listing that was already leased; the bundle did not report them, and no number changes. Thirty-six Gemini episodes were refused when the provider account ran out of credit. The pre-registered bundle keeps them as typed missingness, and the tables use a fill run under its own frozen configuration, whose twelve reruns of completed episodes reproduced their outcomes. On the pre-registered bundle alone (240 Gemini worlds) the leads are +70 [+48, +92] and +80 [+62, +99] under the pre-registered measure and +60 [+39, +81] and +67 [+49, +84] under the corrected one. The corrected measure values a blind signing at the posterior given the inspections and the informative replies. Under it, 21 (GLM), 33 (Luna) and 3 (Gemini) signings change value; Gemini leads by +63 [+43, +84] and +65 [+49, +82] (Holm $p < 10^{-8}$), GLM−Luna is +2 [−20, +24], and each model's difference from the reference becomes −79, −142 and −144. We do not edit the published bundle.

Two CLI agents played the first 20 worlds of the main evaluation set (40 episodes each). Sol 6.1 earns $313 [225, 402] and Opus 5.5 earns $291 [198, 383] a world, against the reference's $331 on those worlds; their difference, +23 [−19, +64], is not statistically established. Both equal the reference where the favourite is sound ($427) and fall short where it is a lemon ($200 and $155 against $235).

This reference is reachable but not optimal. An exact best informed tenant (Appendix C) earns $307 a world on the main evaluation set when it knows the two landlord kinds and how each prices. The v14 models were not told how landlords price, so part of their gap to this tenant reflects missing information rather than poor

judgment. v15 states the pricing rule in its task prompt and uses this tenant as the reference for its inclusion screen and for its gap rows (Section 5.1).

Procurement

The earlier evaluation set. Section 5.1 reports the case's harder evaluation set. The earlier set, which all four models played, has 24 worlds of six kinds. In it, quotes always arrive, the recorded supplier always charges a premium, and newcomers never share a source (Appendix A.2). For a model m with decision loss L_m against the best policy, we report the share of the premium-to-best gap that it closes,

$$G_m = (L_{pay} - L_m) / L_{pay}, \tag{E.1}$$

where L_{pay} is the loss of always paying the premium. G_m equals zero for that rule and one for the best policy, and it is negative for play worse than paying the premium.

Table E.14. The earlier Procurement evaluation set, 24 worlds per model: the share of the best policy's value (it earns $451 a world) that each model gives up and keeps, with 95% intervals over worlds. Always paying the premium, which never tests a newcomer, already keeps 83.3% (it gives up $75.19 a world), so the last row reports the decision itself: the share of the gap between that rule and the best policy that each model closes. On 16 October we reproduced all 96 losses and these intervals with a source-pinned derivation from replay-audited sealed actions; decision loss is separate from the original receipt scores.

	Sol 6.1 (Codex)	Opus 5.5 (Claude Code)	Gemini 3.8 Flash	GPT-6 Luna
Share of the best policy's value given up	0.2% [0.0, 0.5]	1.2% [0.3, 2.4]	16.7% [10.8, 22.5]	16.5% [10.4, 22.8]
Share kept (always paying the premium: 83.3%)	99.8%	98.8%	83.3%	83.5%
Share of the premium-to-best gap closed	98.6% [97.2, 99.7]	92.8% [85.9, 98.4]	0.0% [−34.9, 35.2]	1.1% [−36.6, 37.9]

The frontier agents, run through their CLIs, are at the ceiling. Most of what they lost on a first run came from two rules that the task prompt left unstated: that a quote is final, and that the average is the criterion. This ceiling shows less than it appears to. On this evaluation set a rule that looks one test ahead comes within 0.01% of the best policy's value, so near-perfect play shows that the agents avoid the listed mistakes, not that they plan further ahead than one test. A v2 gate check and pilot (50 episodes, $0.22) found little newcomer testing and required 326 to 3,464 worlds against a cap of 96, so no confirmatory run followed (Appendix F.4); the harder evaluation set of Section 5.1 followed instead. The flash models give up as much as always paying the premium does: their loss arises in the worlds where a newcomer was worth testing.

Further runs on the harder evaluation set (Table E.15). Section 5.1 reports the 20 pre-registered worlds. On them, Sol opens with the kind-blind shortcut's move in 2 of 40 plays, tests a second newcomer in 7 of 40 and looks up a source in 4, so the moves the case is built around are rare in its play. The runs below are a development pilot and three diagnostics, each pre-registered before its plays; none of them tests a claim. These runs, like the pre-registered test, used the case's environment from a development branch, which is now on the main branch with its registration, and every published receipt replays through the runtime's command line (Appendix F.1).

Table E.15. Procurement's harder evaluation set: runs beyond Table 5.3. Shares are of the gap between paying the premium and the best policy, as means over worlds (each world the mean of its plays).

Run	Worlds and plays	Sol 6.1	Opus 5.5	Exact rules on the same worlds
Development pilot	6 worlds, two plays each	50% [27, 71] (bootstrap)	0% in all 12 plays	one-newcomer rule 0%, shortcut 8%, two newcomers 84%
The pre-registered worlds at the earlier set's cohort odds (diagnostic)	20 worlds, one play each	59% [19, 100], median 83%; one play deferred and bought nothing	0% in all 20 plays	one-newcomer rule 58%, shortcut 62%, two newcomers 90%
A task prompt corrected for four wording defects (diagnostic)	the 20 pre-registered worlds, two plays each	no change shown: +7.8 points [−5.6, +21.2]	touched no newcomer in 40 plays	as Table 5.3
How likely a newcomer is to be sound once another of its kind has passed or failed a sample, with and without a clean quote of its own first, asked with no play (diagnostic)	3 worlds, 3 questions, asked twice	18 of 18 within 0.05 of the exact value	18 of 18 within 0.05; earlier attempts, whose schema asked for the model's working, were refused	

A further diagnostic, pre-registered before its plays, tried to certify Gate 1's measured headroom with a frozen control that tests, as the standard then required. We chose GPT-6 Luna at high reasoning effort on the pilot worlds because it tested a newcomer in 5 of 12 plays (Gemini 3.8 Flash at high reasoning effort tested in none). On the pre-registered worlds, with three plays each, it tested in 26 of 60 plays but closed −34% of the gap on average, which is worse than paying the premium. Against this control, 3 worlds score the same at every seed and 18 are matched by a screened rule (the kind-blind shortcut in 14, the one-step rule in 4), so only 2 of 20 pass. The control is weaker than the shortcut, which says more about the control than about the worlds. In a comparison that was not pre-registered, Sol's better play of each world beats every screened rule in 11 of 20 worlds, and if its two plays were read as a control, Sol would pass the check in only 9 of 20 worlds. A world that a model fails to exploit is not a world without headroom: every shortcut loses at least $10 to the exact reference in every pre-registered world. The standard therefore now accepts this exact screen where a case has an exact reference, and the evaluation set passes Gate 1 on it (Appendix F.1); the control's result remains reported as a diagnostic.

The pilot's go rule sized a comparison of the two models beyond the cap of 96 worlds, so the pre-registered test concerned Sol alone. On the pilot worlds the kind-blind shortcut closes 8% of the gap against Sol's 50%; on the pre-registered worlds it closes 33% against Sol's 35%. No comparison with the shortcut was pre-registered.

We score the fixed rules exactly over the 24 worlds. A rule that compares each untested newcomer with paying, looking one test ahead and keeping the recorded supplier reachable, gives up $0.06 a world. Testing the best odds once gives up $45.7, always paying $75.2, testing the cheapest newcomer once $81.4, and working up the price list until one listing qualifies $146.1. Under a first task prompt that did not state that a quote is final or that the average is the criterion, the CLI agents gave up $12.2 [4.3, 23.6] (Sol) and $19.4 [6.0, 38.2] (Opus) a world; stating the two rules reduced the loss by $11.1 [3.6, 22.3] and $14.0 [0.4, 33.5]. Under the second task prompt the agents give up nothing in 20, 17, 8 and 7 of the 24 worlds (Sol, Opus, Gemini and Luna respectively). Both CLI agents pay without a test in all eight worlds where paying is right, and the first newcomer each tests is the right one in 15 and 14 of 16 worlds. The record evaluation set is saturated in the same way: Opus solves 18 of 18 worlds and Sol 16. Of the flash models, Gemini never tested a newcomer and Luna tested in two worlds. The two cannot be distinguished (+$0.8 [−3.7, +6.3]); their loss is $112.8 and

$111.0 a world where a test is the best plan, against $0.0 and $1.0 where paying is, and all 48 episodes are valid and replayed ($0.21). GLM 5.3 Flash did not play, because its provider route changed price before any call.

Data center

v2 (Table 5.4). All freeze and run probes were clean, and all 640 negotiations are valid and replayed ($55.41 at list price). With the walk-away point stated, neither model signs below it as client (0 of 80 for either), and the client seat separates nothing (+0.8 points [−3.2, 4.9]). On v2's deal worlds the best informed turn-by-turn policy gives the integrator 63.5% of the value available at the true client type, whereas Sol earned 33.8%, Opus 26.3% and the price-only rule 26.5%. Sol therefore captures about 20% of the room between the rule and that policy. The earlier comparison (v1) is reported at the end of this subsection (Tables E.16 and E.17), and development results are in Tables E.3 and E.7.

Development sets. The counterpart and probing results below use the development sets. Which model sits in which seat changes the pair's total by +3 [−11, +18] thousand dollars a world, and in head-to-head play Sol is ahead of Opus in 51 of 52 worlds in each seat. Table E.3 reports every counterpart, and Appendix B gives each counterpart style's rule and parameters.

Table E.3. Development results: Sol 6.1 minus Opus 5.5 in the same seat, by counterpart ($ thousands per world, 52 deal worlds).

Counterpart	As integrator	As client
The other model	+248 [+206, +290]	+251 [+209, +293]
Firm population	+17 [−10, +44]	+23 [−2, +48]
Soft	+86 [+56, +116]	+95 [+74, +115]
Price only	+112 [+75, +149]	+54 [+36, +71]
Collapse	+88 [+51, +126]	+148 [+117, +179]
Overreach	+32 [+10, +53]	+108 [+72, +144]

Against the firm population, both models earn about what signing the counterpart's first offer earns (a scripted stand-in that does so earns $122k a world as integrator and $114k as client), but by opposite routes: Opus signs the first offer, whereas Sol bargains until the last move and loses deals. Two copies of Sol split the value evenly (integrator minus client −6 [−94, +81] thousand dollars a world).

Adversarial counterparts.

- *Hardball scripts* (overreach, collapse and price only; 684 of 696 negotiations valid; $33.41). The overreach script tests whether a model refuses a deal worse than walking away. As client, Opus signed below its own walk-away point in 23 of 52 worlds and earned −8 thousand dollars a world on average, whereas Sol signed none and earned $100k. The collapse script holds a high ask until its final proposal and then concedes, so it rewards a model that does not settle early: as client, Sol earned $187k a world and Opus $40k. The price-only script tests whether a model sets the clauses itself. As integrator, Opus reached the best clause set in 0 of 52 worlds and lost 15 deals, whereas Sol reached it in 10.
- *A probing counterpart* (20 worlds, with each model in each seat against the prober and against a silent script; 158 of 160 negotiations valid; $7.92). Both sides' task prompts carried a planted confidential name and figure that change no payoff, and the prober's notes asked for the other side's costs and for the confidential item. Neither model disclosed a confidential item in any negotiation. When asked for its own costs, Sol as integrator stated its pre-staging cost in 18 of 19 negotiations, and Opus in none of 40. Nothing in the task prompt says that a side's own costs are confidential, and no model stated a false cost. Asking changed no model's earnings measurably (−15 [−86, +57], +4 [−18, +25], +1 [−25, +26] and +3

[−15, +22] thousand dollars a world); the intervals are wide, so this does not show that disclosure is harmless. Unprompted, across 3,492 notes, no model stated its own private cost. In 114 negotiations a note called a price final or a floor, and in 112 of them the stated number was better for the speaker than its true limit.

A pre-registered test. A look back over these runs suggested two signals: Opus's counter-offer price seemed to follow its private value, and Sol's clause requests seemed to follow its risk charge. We committed the claims, measures, exclusions and six primary predictions before any run, on the fresh worlds of Appendix E.2. Sol asks for 1.82 [1.23, 2.40] more protective clauses (of six) under the high charge (supported, Holm $p < 0.001$), which is 1.58 [0.95, 2.22] more than Opus does. At the true charge, the clauses Sol chose for its own type are worth about +120k [+83k, +156k] dollars a world more in joint value than those it chose for the other type (a secondary measure; joint value is the pair's surplus, not Sol's own earnings). Opus's price pass-through is −0.07 [−0.18, +0.05], so that prediction is not supported; the earlier estimate of 0.52 came from counter-offers alone, compared across worlds, and we withdrew it. The fitted reading counterparts yielded +2k [−14k, +17k] and −8k [−42k, +26k], but their rules came from the confounded look-back data. The counterpart that reads Opus's price used the withdrawn slope of 0.52, so this run tests a misspecified reader rather than a well-fitted reader's ability to exploit Opus. The within-world price result remains unsupported, and the reader-effect estimates cannot establish that disclosure is harmless.

Of the 2,456 negotiations, 2,417 are valid. Every failure is a CLI service error. The original bundles keep the failures as missing, and we re-ran 16 of them under separate frozen configurations by an explicit, recorded decision.

The earlier pre-registered comparison (v1). Before v2, we pre-registered a comparison on 60 included worlds (from 2,729 scanned), 45 of them with a mutually beneficial deal. It differs from v2 in three ways: the two command-line tools were not equally isolated, the client's task prompt did not state its walk-away point, and the worlds were not screened for the integrator's headroom over a price-only rule. We do not pool v1 and v2.

Table E.16. Data center v1 (14 October): Sol 6.1 (Codex) and Opus 5.5 (Claude Code). Differences in a seat's earnings above its outside option, at true costs, on the 45 deal worlds, in percentage points of the value available in a deal ($322k a world); intervals are 95%, with Holm correction across the campaign's primary contrasts. The campaign has eight documented departures from its plan and a partial Gate 5 verdict (Appendix F.5).

Primary contrast	Worlds	Estimate [95% interval]	Holm p
Head to head: Sol minus Opus earnings, seat-averaged	45 deal	+59.1 [46.0, 72.1]	3.2e-11
Firm population: Sol minus Opus earnings, seat-averaged	45 deal	+8.2 [3.3, 13.2]	0.0017
Client signs below its walk-away point: Opus / Sol	all 60	45% vs 5% (27 and 3 of 60)	2.4e-7

Sol's lead against the firm population is detectable but smaller than the plan's smallest meaningful difference of 15.5 points ($50k; the 98.3% interval is [2.1, 14.3]). Across v1's pilot and three arms, 1,175 of 1,176 negotiations are valid ($74.41 at list price). The two tools were not equally isolated: Codex's context listed eight of the operator's own skills, two of them described in benchmark and evaluation terms, whereas Claude Code's context listed none (Table H.1). Every v1 contrast between Sol and Opus carries that difference. Once v2 stated the walk-away point, neither model signed below it as client.

Against the firm population, redline fees were the largest part of v1's joint loss (69.9% of Sol's $103.4k a world and 40.3% of Opus's $70.8k, over all 60 worlds with seats averaged). Opus's next largest part (29.5%) was harmful agreements: in the 15 worlds where no agreement beats both outside options, it signed anyway. Table

E.17 gives v1's joint loss by pairing, and Appendix E.9 replays it at other fee levels.

Table E.17. Data center v1: joint loss by pairing ($ thousands per world, all 60 worlds).

Integrator / client	Inferior clauses	Harmful agreement	Missed deal	Redline fees	Total
Opus / Sol	9.3	1.2	3.7	63.3	77.5
Sol / Opus	22.6	34.8	25.2	44.2	126.8
Opus / firm	2.2	0.0	12.2	30.0	44.4
Sol / firm	1.1	0.0	29.1	60.8	91.0
Firm / Opus	15.5	41.8	12.9	27.1	97.2
Firm / Sol	14.3	4.0	13.8	83.8	115.8

E.5 Failure classes: rules and full tables

Table E.4. The shared classes.

Class	What happened
Information not acquired	acted without information worth acquiring: signed an uninspected listing whose quality nothing it knew had settled; paid the premium or awarded the recorded supplier where the best policy tested a newcomer; decided before a required fact
Acted contrary to information held	the trajectory contains information (an inspection result, a landlord's reply, a sample result, revealed facts) under which the reference acts differently, and the subject did not: signed a hold that, under the landlords' pricing rule, only a lemon's landlord offers, at a rent above a lemon's value; awarded a supplier its own test showed bad; stopped after one bad newcomer where another test was worth it; tested the wrong newcomer or in the wrong order; a decision the answer key contradicts after every required fact was revealed. The class records that the information was in the trajectory, not that the subject could interpret it
Worse terms	agreed to a worse deal than one available, with no protocol error: a verified listing worth less net of rent than the reference's; a worse award on a qualified supplier; a clause set worth less to the pair than the best one
Commitment error	signed what it should have refused or refused what it should have signed: no lease where the reference leased; deferred where an award or a test was worth more; no deal where one was available
Interaction cost	the charge for moves the case prices: redline fees after the opening; inquiries, counter-offers and checks; inspection spend above the reference's on a lease of the same value. Any gain such a move brings appears in another class (a better clause set is smaller worse-terms loss), so this class prices the moves and does not judge them
Procedure and format	an action the environment rejected or could not parse; a fact request the case rejects; a format the verifier rejects
Operational	an episode with no measurement; counted, never scored

For a unit u (an episode, an action or a part of a grade, depending on the case), the shortfall is the reference's value minus the agent's, and the rules assign it to the classes in priority order:

$$S(u) = V_{ref}(u) - V(u) = \Sigma_c\ S_c(u), \tag{E.2}$$

where c ranges over the six scored classes and each part of the loss goes to the first class whose rule it meets. The priority order makes the decomposition an accounting identity, not a causal attribution.

Rules per case. *Housing.* The unit is the episode (one world under one landlord kind). The shortfall is the reachable reference's net minus the tenant's net under v15's primary measure (the reply-conditioned measure), and the rules apply in the order procedure, contrary to information held, information not acquired, commitment error, worse terms and interaction cost. A known lemon signed at a rent that its lemon value covers is judged on terms, because the reference does the same. Signing a revealed lemon counts as acting contrary to

information held: v15's task prompt states how landlords price, so the acceptance that reveals a lemon follows from a stated rule. The rules are v14's. They read seat 0's decisions from v15's seat-0 receipts, the reference is `inspect_lowball` scored as v15's scorer scores it, and every episode's gap reproduces the published v15 analysis. v14's class totals (Gemini $78.9, GLM $142.3 and Luna $144.3 a world) remain in the source package. *Procurement.* The unit is the action. Each action's loss against the best policy (these losses sum to the decision loss) goes to a class according to what the action did. A request to the recorded supplier counts as information not acquired only when the episode never tests a newcomer, and as acting contrary to information held otherwise. *Data center.* The unit is each part of a negotiation's grade. No deal is a commitment error, labelled by the side that walked away. Allocation loss in a world with positive available surplus counts as worse terms, while allocation loss in a world without positive available surplus is a harmful agreement, which is a commitment error. Redline fees are interaction cost. Figure 5.2c separates harmful agreements from missed beneficial deals. We count signing below one's own walk-away value separately: that individual loss is not an additive component of joint loss, although the same agreement may also destroy joint value. *Refund.* The parts of the published gap decomposition map onto the classes: deciding before a required fact maps to information not acquired, a wrong decision to acting contrary to information held, and re-asked or unknown facts and the null refund method to procedure and format.

Interpreting negotiation costs. A redline that improves the final agreement's joint value by $100k while costing $25k reduces total shortfall by $75k: allocation loss falls by $100k and interaction cost rises by $25k. The score therefore credits improvements net of their cost. The full-information reference knows both private types and can propose the best agreement in the free opening move. Its advantage includes negotiation costs that agents with limited information may need to incur; the decomposition does not identify which costs were avoidable. It also does not establish robustness to the chosen $25k fee per additional redline. Appendix E.9 tests scoring sensitivity by replaying the same actions at alternative fees; measuring behavioural adaptation would require new model runs under pre-registered fee settings.

Table E.5. Housing and Procurement, every subject ($ per world; Housing v15 on the 260 main worlds of Table 5.2, GLM, Sol and Opus from their pre-registered addenda; Procurement on the 20 pre-registered worlds of Table 5.3, each world the mean of its two plays).

Class	Housing Gemini	Housing Luna	Housing Sol (addendum B)	Housing Opus (addendum B)	Housing GLM (addendum A)	Proc. Sol	Proc. Opus	Proc. Gemini	Proc. Luna
Information not acquired	−0.2	14.9	0	1.4	18.9	17.5	51.0	49.4	55.4
Acted contrary to information held	1.9	22.2	0	0	49.6	13.9	0	0	0
Worse terms	40.7	54.3	15.7	38.8	40.1	0	0	0	0.5
Commitment error	8.9	8.6	−0.0	6.3	7.2	0	0	0	0
Interaction cost	−0.3	−0.1	−0.4	−0.4	−0.2	1.1	0.2	1.9	2.2
Procedure and format	0	0	0	0	11.7	0	0	0	0
Total	51.1	99.9	15.3	46.1	127.3	32.5	51.2	51.2	58.1

Housing's totals reproduce the published v15 differences from `inspect_lowball` (51.13 and 99.92 $ a world; GLM's addendum row R3, 127.32; Sol's and Opus's addendum row R, 15.34 and 46.07), and every one of the 520 main episodes per model completed, so no world is excluded. Main-world means average both landlord arms within a world. We retain negative amounts where the model beats the reference. The inspection-cost row is a residual after the earlier classes, so inspection costs already inside other classes are not counted again.

Table E.6. Data center development ($k joint value lost per deal world) and Refund (points of episodes unresolved).

Class	DC Sol int. × Opus	DC Opus int. × Sol	DC Sol × firm	DC Opus × firm	Refund DeepSeek	Refund Gemini 2.5 FL	Refund GPT-5.6 Luna	Refund Grok
Information not acquired	—	—	—	—	44.2	0	15.0	29.2
Acted contrary to information held	—	—	—	—	0	0	0	0.8
Worse terms	35.4	12.2	9.1	4.8	0	0	0	0
Commitment error	0	0	32.0	6.9	—	—	—	—
Interaction cost	46.2	66.3	80.2	29.8	—	—	—	—
Procedure and format	0	0	0	0	5.8	98.3	5.8	17.5
Total	81.5	78.6	121.3	41.5	50.0	98.3	20.8	47.5

In these development runs, 7 of Sol's 8 lost deals against the firm script occurred when the script walked away from Sol's last offer. Signing below one's own walk-away point, which the table does not include, occurred for Opus as client in 12 of 52 worlds against Sol (in 4 of them Opus proposed the losing terms itself) and in 2 against the firm script, and for Sol as client in 1 against the firm script; neither model did so as integrator.

E.6 Two frontier agents across three cases

Table E.7. Development comparisons of Opus 5.5 and Sol 6.1 on the three cases both played; Data-center confirmatory results are in Table 5.4 (v2) and Table E.16 (v1).

Case (worlds)	Measure	Opus 5.5	Sol 6.1	Sol relative to Opus	Reference
Housing (20)	tenant net per world	291 [198, 383]	313 [225, 402]	+23 [−19, +64]	scripted reference 331
Housing, favourite a lemon (10)	tenant net per world	155	200	not tested	scripted reference 235
Procurement pay-or-test (24)	given up per world	5.4 [1.2, 10.6]	1.0 [0.2, 2.1]	4.4 [0.6, 9.0] less given up	best policy gives up 0
Procurement, record evaluation set (18)	worlds solved	18	16	regret −20 [−48, +7] (Opus lower)	informed policy 18
Data center development, firm population (52)	earns, integrator / client ($k)	122 / 107	139 / 130	+17 [−10, +44] / +23 [−2, +48]	first-offer stand-in 122 / 114
Data center development, head to head (52)	earns, integrator / client ($k)	60 / 25	308 / 276	+248 / +251	

E.7 Current shortfall panels: scope and uncertainty

Figure 5.2 is descriptive accounting computed after the runs, not an additional pre-registered test. Its Data-center panel uses all 80 included worlds of v2, including 8 with no positive available surplus, whereas Table 5.4's earnings contrasts use the 72 worlds with a deal. Both focal roles receive equal weight within each world, after the integrator's two plays against the firm client are averaged. We do not add individual signings below the walk-away point to joint loss: as integrator against the firm client, Opus signed below its outside option in 2 of 160 plays and Sol in none, and as client neither model did so in 80. Table E.6 preserves the earlier development decomposition on its 52 deal worlds, and Table E.17 gives v1's decomposition by pairing.

Table E.8. Data-center joint loss by pairing in v2 ($ thousands per world, all 80 worlds; the integrator's two plays against the firm client averaged within each world).

Integrator / client	Inferior clauses	Harmful agreement	Missed deal	Redline fees	Total
Opus / Sol	12.6	0.0	24.3	74.4	111.4
Sol / Opus	22.4	0.0	26.1	68.1	116.7
Opus / firm	2.2	0.0	9.1	40.0	51.3
Sol / firm	1.7	0.0	45.9	73.9	121.6
Firm / Opus	17.0	0.0	48.3	50.6	115.9
Firm / Sol	12.4	0.0	53.6	92.8	158.8

Table E.9. Distribution of total shortfall for the subjects in Figure 5.2 (each case's own units).

Case / subject	Units (n)	Mean [95% interval]	Median [Q1, Q3]	Zero / negative units
Housing: Gemini 3.8 Flash	260	51.1 [38.3, 63.9]	29.9 [0.0, 98.3]	60 / 45
Housing: GPT-6 Luna	260	99.9 [83.8, 116.0]	72.1 [0.0, 167.6]	46 / 32
Housing: Sol 6.1 (addendum B)	260	15.3 [6.2, 24.5]	0.0 [0.0, 53.7]	78 / 59
Housing: Opus 5.5 (addendum B)	260	46.1 [33.9, 58.2]	27.8 [0.0, 96.0]	61 / 47
Housing: GLM 5.3 Flash (addendum A)	260	127.3 [105.2, 149.4]	79.9 [0.0, 212.8]	36 / 39
Procurement: Sol 6.1	20	32.5 [23.8, 41.3]	33.3 [21.5, 48.7]	0 / 0
Procurement: Opus 5.5	20	51.2 [46.3, 56.2]	50.2 [45.1, 57.7]	0 / 0
Procurement: Gemini 3.8 Flash	20	51.2 [46.3, 56.2]	50.2 [45.1, 57.7]	0 / 0
Procurement: GPT-6 Luna	20	58.1 [52.3, 63.9]	57.2 [50.4, 67.1]	0 / 0
Data center: Opus 5.5	80	83.6 [68.8, 98.4]	57.2 [27.8, 137.3]	0 / 0
Data center: Sol 6.1	80	140.2 [120.6, 159.7]	117.3 [87.5, 189.8]	4 / 0
Refund: GPT-5.6 Luna	6	20.8 [-1.0, 42.7]	20.0 [3.8, 28.8]	2 / 0
Refund: Gemini 2.5 Flash Lite	6	98.3 [94.0, 102.6]	100.0 [100.0, 100.0]	0 / 0

Opus and Gemini bought from the recorded supplier in every play, so their Procurement rows coincide. The intervals are Student t intervals over independent worlds (for Housing after averaging the arms, and for Data center after averaging the focal roles), or over the six fixed Refund cases after averaging 20 episodes per case. They describe variation in these panels; they do not establish a model ranking or coverage of a field population. They are descriptive intervals that we derived afterwards, not replacements for the campaigns' pre-registered analyses. A Refund interval can exceed the score's 0–100 bounds; we leave it untruncated to expose the weakness of an approximation with six fixed cases. Housing has one retained run per world and landlord arm,

so its run-to-run variance is unmeasured. Procurement and Data center have two plays per world: the two plays of a world scored identically in 7 to 20 of the 20 Procurement worlds, depending on the model, and ended identically in 47 (Opus) and 28 (Codex) of the 80 Data center worlds (Appendix F.1). Refund has repeated episodes within fixed cases, and the derived evidence retains its per-case outcome distributions. The source package includes component-level intervals, distributions, per-world amounts and input digests, together with a reproduction script that makes no provider call.

E.8 Exact values for Figure 5.2

The figure draws the estimates below. Each component keeps its case's native units and classification rules, and the figure adds no new analysis or cross-case aggregate.

Table E.10. Shortfall by case. Housing uses v15 (Table 5.2) under its primary measure; Data center uses v2, the comparison of Table 5.4; Procurement uses the 20 pre-registered worlds of Table 5.3. Refund retains its development panel.

(a) Housing

Dollars per main world below the reachable scripted reference (`inspect_lowball`), under v15's measure v2.

Component	Gemini 3.8 Flash	GPT-6 Luna	Sol 6.1 (Codex, addendum B)	Opus 5.5 (Claude Code, addendum B)	GLM 5.3 Flash (addendum A)
Information not acquired	−0.2	14.9	0.0	1.4	18.9
Contrary to held information	1.9	22.2	0.0	0.0	49.6
Worse lease terms	40.7	54.3	15.7	38.8	40.1
Lease commitment error	8.9	8.6	−0.0	6.3	7.2
Procedure or format	0.0	0.0	0.0	0.0	11.7
Inspection cost	−0.3	−0.1	−0.4	−0.4	−0.2
Total shortfall	**51.1**	**99.9**	**15.3**	**46.1**	**127.3**

The two landlord arms are averaged within each of the 260 main worlds, and no episode is missing. GLM (addendum A) and the two command-line agents (addendum B) played in pre-registered addenda outside v15's contrasts. GLM's procedure losses are actions the environment refused, chiefly malformed commits ($5.7 a world, 14 episodes) and offers on listings already taken ($2.1, 5 episodes). Negative amounts mean that the model did better than the reference in that class; they are not drawn in Figure 5.2.

(b) Procurement

Dollars per world below the best policy given the available information, on the 20 pre-registered worlds of Table 5.3 with two plays each.

Component	Sol 6.1	Opus 5.5	Gemini 3.8 Flash	GPT-6 Luna
Information not acquired	17.5	51.0	49.4	55.4
Contrary to held information	13.9	0.0	0.0	0.0
Worse terms	0.0	0.0	0.0	0.5
Award checks	1.1	0.2	1.9	2.2
Total shortfall	**32.5**	**51.2**	**51.2**	**58.1**

The panel shows all four models, with each world the mean of its two plays. Sol runs through Codex, Opus through Claude Code, and Gemini and Luna through OpenRouter. We re-derived the decision loss from every play's replayed trace and matched every world against the read-outs. The pre-registered rules file an award's whole loss under information not acquired when the award came before a worthwhile newcomer test, including the part caused by a worse quantity. For Luna, which ordered 50 units where 55 was best in 15 of 40 plays, that part is $6.33 a world; if the loss were split into its timing and terms parts, Luna's worse-terms loss would be $6.85, its whole loss beyond paying the premium. The other three models made no such awards.

(c) Data center

Thousands of dollars of joint value lost relative to the full-information bound, in v2 against the firm counterparts.

Component	Opus 5.5	Sol 6.1
Inferior contract clauses	9.6	7.1
Harmful agreement	0.0	0.0
Missed beneficial deal	28.7	49.7
Redline fees	45.3	83.4
Total shortfall	**83.6**	**140.2**

All 80 worlds, with both focal roles averaged within each world. Joint outcomes reflect both parties' actions.

(d) Refund

Percentage points below full resolution under the verifier, in V2.1.

Component	GPT-5.6 Luna	Gemini 2.5 Flash Lite
Required facts skipped	15.0	0.0
Procedure or format	5.8	98.3
Total shortfall	**20.8**	**98.3**

120 episodes per model, averaged within each of the six fixed cases and then across cases.

We omit rows that are zero for every displayed model, and rounded components may not add up to rounded totals. Signing below one's walk-away value is a separate, individual outcome. In v2 neither model signed below its walk-away value as client, whether against the firm integrator or head to head (0 of 80 each); as integrator against the firm client, Opus signed below its outside option in 2 of 160 plays

and Sol in none. Table E.16 gives v1's counts.

E.9 Negotiation fees in v1: fixed-action sensitivity

We replayed all 240 valid firm-population negotiations of v1 (Table E.16; 60 worlds, two models and two focal seats) at six fees per paid redline: $0, $5k, $12.5k, $25k, $50k and $100k. All original final states and six score leaves matched the saved states and receipts. The 1,440 fee-specific trajectories used the environment's own transition and scoring functions and held offers, clauses, notes, signatures and walk-aways fixed. This is an exploratory analysis performed after the runs, not another pre-registered test or a model rerun. The opening move remains free of fees. v2 pre-registered a smaller replay of the same kind (Section 5.2).

The two samples answer different questions. v1's joint loss covers all 60 worlds, whereas its pre-registered earnings contrast (Table E.16) uses the 45 worlds with positive available joint surplus. "Deal-feasible" describes those worlds, including negotiations in them that ended without agreement; it does not select only signed contracts. Means first average both focal seats within a world and then average across worlds, so the independent unit is the world, not each correlated seat observation.

Table E.11. Fee sensitivity on the 45 deal-feasible worlds ($ thousands per world; fixed actions). Sol earnings lead is Sol minus Opus against the firm population. Lower joint loss is better; a positive earnings lead favors Sol.

Fee per redline	Sol joint loss	Opus joint loss	Sol earnings lead
0	38.9	28.5	+54.8
5	56.0	35.6	+49.1
12.5	81.7	46.2	+40.6
25	124.4	63.8	+26.4
50	210.0	99.1	-1.9
100	381.1	169.6	-58.6

On these 45 worlds, fees amplify Sol's disadvantage in joint loss but do not create it: Opus retains the lower loss at zero fees. Sol's own earnings lead, by contrast, grows when fees are removed. Because behaviour is held fixed, each model's earnings are a linear function of the fee, and the two models' earnings are equal at approximately $48.3k per redline. This threshold describes the recorded actions; it is neither an optimal fee nor a prediction of how either agent would respond to a different fee.

Table E.12. Joint-loss sensitivity across all 60 worlds ($ thousands per world; fixed actions). This is v1's full population, including 15 worlds without positive available joint surplus.

Fee per redline	Sol joint loss	Opus joint loss
0	31.1	42.3
5	45.6	48.0
12.5	67.3	56.6
25	103.4	70.8
50	175.7	99.4
100	320.3	156.5

Across all 60 worlds, fees contribute 69.9% of Sol's and 40.3% of Opus's total joint loss at $25k per redline. On this population the joint-loss ranking reverses without fees, with equality near $6.4k per redline, so the 45-world result cannot establish a fee-independent ranking for the full evaluation set. On the 45-world subset, the corresponding fee shares are 68.8% and 55.3%.

Useful changes and limits. Among the paid redlines that the models authored in the 45 deal-feasible worlds, 23 of Sol's 150 and 5 of Opus's 48 increased the immediate full-information clause value by more than the $25k fee. One Sol redline improved clause value by $87.5k, a net improvement of $62.5k. These are local comparisons: an edit may later be undone, and a negotiation may remain unsigned. This count of clause value does not capture price concessions or the information value of negotiating. The fee shares include both parties' fees, whereas these counts include only the focal model's edits. The full-information reference keeps its advantage, and this analysis neither validates the fee's empirical calibration nor measures adaptation to it.

The source package retains the supplied analysis, its reproduced outputs, its input hashes and the specific redline discussed above. The reproduction uses the retained campaign logs, which the compact package does not include in full.

Appendix F. Quality control in detail

This appendix describes only the checks that we built and applied to the four cases, with two caveats that apply throughout. First, the initial quality-control profiles and reference-run tests were written after development; only the agreement's confirmatory run followed its profile and a dry run that made no provider call. Second, we do not describe rules of the quality-control standard that were not applied to these cases.

F.1 Profiles and gates

The quality-control standard defines six quality-control gates, together with the rule that a change to any bound input reopens them, and a repository test rejects a registered family that has no profile. Each of the four current designs now has a section in its family's profile with a typed status and a blocker for each gate (Table F.1), and none passes Gates 1 to 5. Housing v14 is `partial` throughout (v15, discussed below, passes four of the five). Procurement's harder evaluation set, used in Table 5.3, passes Gates 0, 1, 2, 3 and 5 and is `partial` at Gate 4 (profile section 8.2). Its environment and its registration with the runtime reached the main branch on 20 October, the registration by the owner's override without a review by someone other than its author. Its 12 published bundles replay through the runtime's command line, 432 of 432 receipts, among them 100 from five scripted baselines that played the pre-registered worlds with no model, each scoring exactly as the reference scores its moves. Gate 4 stays `partial` because the command-line agents take no seed. Gate 1 passes on the exact-reference route that the standard added on 21 October: the reference is exact, and the six shortcuts that the profile names (paying the premium, three fixed testing rules, the one-step rule and the kind-blind shortcut, the last two found in earlier versions) each lose at least $10 in every pre-registered world. Rules that do make the decision come closer: looking two newcomers ahead loses $8.38 a world, and as little as $0.09 in one world. Before the route existed, we tried to establish measured headroom with a frozen control that tests, pre-registered before its plays. The control tested but lost to the kind-blind shortcut, so by that route only 2 of the 20 pre-registered worlds passed (Appendix E.4). A control weaker than the shortcuts measures itself rather than the worlds, which is why the standard now uses the exact reference where a case has one. The models' two plays of a world scored identically in all 20 worlds for Opus and Gemini and in 7 for Sol, so the effective sample is the 20 worlds, as the world-mean intervals assume. The earlier pay-or-test v1 evaluation set (Table E.14) fails Gates 1 and 3. v2 passes the scripted Gate 3 checks and its screen meets the exact-reference route, but its pilot stopped at sizing and the case moved to v3 (Appendix F.4). The agreement's development sets fail Gate 1 because inclusion and type assignment were confounded. Its first confirmatory run (v1) used the included evaluation set, which is still `partial` at Gate 1 because of the client seat's construct limit and incomplete measured-headroom qualification (Appendix F.2). Refund V2.5 fails Gate 3 on its lower controls (Appendix F.4) and is `partial` at Gates 1, 2 and 4: a constant full refund solves 38 of its 180 policy cases, its schema accepts a null refund method that its verifier then fails, and no adapter admits a model profile. No model has played it, so Gate 5 has not run.

We pre-registered and ran Housing v15 after these profiles were first written. It passes Gates 1, 2, 3 and 5 and is `partial` at Gate 4: - **Gate 1.** Its evaluation set is screened for headroom against `inspect_then_sign`, and its holdout changes a mechanism. Scored per world afterwards, every rule that skips the decision loses at least $25 to the best informed tenant in every world (the smallest losses are $28.06 for never trading, $92.37 for signing anything, and $25.69 for inspecting and then signing at the asking rent). `inspect_lowball`, which inspects the favourite, offers $0 and signs the counter-offer when the inspected value covers it, comes within $25 of the reference in 229 of 260 worlds and equals it in all 60 worlds with a sound favourite. The headroom this evaluation set leaves therefore lies in carrying out a simple procedure, and every model falls short of that rule on average (Table 5.2). - **Gate 2.** We sealed a bundle of golden receipts through its builder, and all 1,720 episodes replay. - **Gate 3.** Its beatability rule was pre-registered before any result, both models pass it, and its receipts score the tenant who decides. - **Gate 5.** Its sizing record and its A/A arm preceded the freeze of the configuration. - **Gate 4, partial.** The pre-registered one-call probe of each model profile ran only after the freeze. It found structured output working and both reasoning tiers in effect: every sealed call's raw response carries the provider's count of reasoning tokens, which the kernel's OpenRouter client does not copy into its own field.

The agreement's v2 confirmatory run (Table 5.4) passes Gate 3 for its integrator seat: the price-only rule leaves headroom in every deal world by construction, the best informed integrator earns far more than either model, and the evaluation set separates the models. It is `partial` at Gates 0, 1, 2, 4 and 5. Its registry entry awaits a review by someone other than its author and its code is not yet on the main branch, so replay through the runtime's command line has not run. Its grid reference is a lower bound, so the exact-reference route does not apply, and a frozen control played on the already frozen evaluation set could only count the worlds that pass. Gate 1 can therefore pass only on a new evaluation set with an exact reference and a shortcut screen in every world; the client seat also retains its construct limit. Scoring every scripted integrator rule against the grid reference shows what a new set must remove: a shortcut comes within $25k of it in 12 of the 80 worlds, namely all 8 in which no deal is possible (walking away earns the reference, and every scripted rule walks away) and 4 of the 72 deal worlds, three of which have a total ceiling below $25k. The shortcuts still lose $104.8k to $205.2k per deal world on average, so the evaluation set separates the models in aggregate. A model's two plays of a world ended identically in 47 of 80 worlds for Opus and 28 for Codex, and some fixed negotiating style comes within $25k of the reference in 27 of the 72 deal worlds. Through the environment's hooks, the full-information pair loses nothing in any of its 80 worlds.

We applied the rule that a changed control requires a new frozen configuration: Housing's 36 refused episodes ran later under their own configuration, and Procurement's corrected schema ran under new configurations.

Table F.1. Benchmark QC gates.

Gate	Question	Stops when	In the four cases
0. Profile	Is there a place to record each gate's verdict?	the family has no QC profile	an earlier Procurement design's construct check found a public-price rule beating the model by $28.50 a world; with no profile, the verdict was read as a campaign finding and lost. The profiles followed initial development; the agreement's confirmatory run followed its profile

Gate	Question	Stops when	In the four cases
1. Task distribution	Are the worlds valid, distinct, split and informative?	an unverified bound, duplicates across clusters, development and holdout overlap, no measured headroom	Housing v14 includes a seed on the tenant's own structure and reads quality afterwards, with a holdout from a disjoint seed range; pay-or-test v1 fails: a rule reading only public information solves 23 of 24 worlds; v2's screen meets the exact-reference route, but its 50-episode pilot found almost no newcomer testing and stopped at sizing; the pre-registered v3 evaluation set passes on that route, every named shortcut losing at least $10 in every world
2. Environment and verifier	Do actions change the intended state, and is every score rebuilt from sealed evidence?	invalid actions mutate state, an oracle disagrees, hidden state leaks, replay differs	Housing v14 has 1,764 scored completions and 36 failure exclusions: completed receipt scores replay from sealed events, and failure rows are verified as exclusions; the headline measure is computed outside the receipts. Procurement's harder evaluation set passes: its 12 published bundles replay through the runtime's command line, 432 of 432 receipts
3. Construct and baselines	Does a better-informed policy do better, and do shortcuts lose?	stronger and weaker policies indistinguishable, a shortcut drives the score	pay-or-test v1 fails: Procurement's fixed rules give up $45.7 to $146.1 a world, but a one-step comparison reaches the best policy to within $0.06; v2, built after the runs, passes for scripted policies: every screened rule, the one-step rule included, gives up at least $10 in every world; Refund V2.5 fails: under a rule committed before its controls ran, naive controls come too close to the reference in three of four seats
4. Attribution and controls	Does the result support the attribution it claims?	a control that was not pre-registered changes, the matrix is incomplete, the block cannot carry the claim	the three v14 frozen configurations differ in model and route and also in reasoning tier, sampling and cost ceiling, a choice recorded in the configurations but not stated in the contract; data-center claims name their counterparts
5. Confirmatory reliability and publication	Was the comparison sized, frozen and executed without tuning on outcomes?	incomplete pilot, a frozen control changed, selective missingness, a guardrail no behaviour can fail	Housing v14 has no sealed sizing record. The agreement's confirmatory run completed pilot, sizing, freeze and execution; its status is `partial` pending closure of analysis-digest and exclusion-verification discrepancies (F.5). Procurement's pre-registered test passes: its plan preceded every play, it ran as pre-registered, and its bundles are published and replay

F.2 Checks applied to the four cases

Table F.2. What each kind of verifier must be shown to do, and how the four cases show it.

Kind	Must be shown	How it is shown in the four cases
Answer	the key is correct and complete for every included case	Refund: inclusion requires the policy's public compiler to reproduce the hand-written reference resolution on every case
Rule	every rule is stated to the agent, and each check fails on a counterexample	Procurement tests most of the rules its task prompt states against the environment; the agreement case tests its stated walk-away cost; two checks (the agreement's oracle grade and Housing's ceiling) have mutation tests; a rule left unstated or a schema that disagrees with its parser is a defect
Value	the reference agrees with the environment, and a simple rule does not already match it	each reference is played on every world (below); Procurement computed its one-step rule before play, and that rule matches its best policy

Kind	Must be shown	How it is shown in the four cases
Relative	the counterparts are named and the gap is reported per counterpart	the data-center counterpart table (Table E.3)
Rubric	the rater agrees with people and resists gaming	not used

- **Regeneration.** In all four cases a test reproduces every committed evaluation set byte for byte from its generator and seeds.
- **Reference run on every world.** Tests written after initial development play each reference through the real environment and verifier with no model. The Housing reference and a full-information tenant play all 600 v14 episodes, reproducing the published per-episode values and reaching the ceiling in every episode (offline, against the same functions that wrote the bundle). Procurement's best policy equals the environment's average over the 27 type profiles in all 24 worlds of the earlier evaluation set. On the harder evaluation set, the best policy and the one-step rule equal their averages over 400 plays through the environment, within four standard errors, on all 12 pilot worlds and all 20 worlds of the pre-registered test; we checked those 20 worlds on 20 October, after the runs. The Housing full-information tenant reaches the ceiling on all 320 v15 worlds. The data-center oracle pair loses nothing in all 116 worlds of both development sets, against an optimum that the test computes independently, in all 60 worlds of v1's evaluation set and in all 80 of v2's; we checked v2's worlds on 20 October, after its runs. A scripted dry run of 120 negotiations covered all 60 included agreement worlds before the confirmatory run. Refund V2.5's release requires the reference to receive full credit on every case. The initial tests were written after development.
- **Replay.** The kernel's replay recomputes each receipt's own score from its sealed events without any provider call. It covers the 1,764 completed Housing v14 receipts (the other 36 rows are verified exclusions for operational failure), the development agreement receipts, which were replayed in-line, and all 1,295 scored receipts from the agreement's dry run, pilot and three confirmatory arms, which were replayed offline. We checked one failure receipt separately for its digest and seal, because the pinned driver cannot classify exclusions. Replay also covers the procurement panels' receipts, replayed in-line, and Refund V2.5's 816 trajectories, which make no provider call (they are unpublished, and 174 of them are distinct). The replay certificate does not cover two headline measures. Housing's seat-0 measure is computed by case code over each episode's outcome, whereas the receipts' own scores total all six tenants; the derived analyses of Table 5.2 re-derived the measure from the sealed event logs, and we re-extracted and regenerated it exactly on 16 October. Procurement's original report read an unsealed action list through an unpinned reference. The reanalysis of 16 October audits all 144 receipts (96 under the final task prompt and 48 under the earlier one), reads the parsed actions from their sealed logs, and pins the extraction, the reference, the report and the case inputs. Every per-world loss and the intervals of Table E.14 reproduce. This is a separately verified derivation; we have not inserted decision loss into the original receipts. The Housing receipts also name the model for the five tenant seats that a script played; the true seat map is in an unsealed log.
- **Inclusion before types.** After development and before the confirmatory run, the data-center agreement gained an evaluation set filtered by its own inclusion criteria: 60 worlds from 2,729 scanned, each included on its public parameters and the type prior before its hidden charge exists, and then given types drawn independently of its design cell (χ^2 p 0.46; the same test rejects the pattern of the development sets). Tests play every shortcut through the environment for every pair of types and confirm the loss that the inclusion screen assigned, and the oracle pair loses nothing in all 60 worlds. A separate 12-world pilot and the 60-world confirmatory run (v1) have been completed; the pilot supplied the sealed sizing record. v2's evaluation set applies the same criteria to fresh seeds and adds a headroom screen against one control,

the price-only rule (80 worlds included of 5,436 scanned). This is evidence of play and variance, not a completed headroom verdict across all controls. The client seat still fails the dependence-on-counterpart-type rule.

- **Visibility.** In Housing, Procurement and Refund, tests check that named hidden fields are absent from the observations; the agreement case has no such test.
- **Paired plans.** A Housing test rebuilds the six v14 plans that ran, matches their identifiers, and asserts that they differ only in model and route, reasoning tier, sampling and cost ceiling, the last three being choices that the contract did not state. Refund tests that a plan has a single agent seat, and the agreement case tests its evaluation sets.
- **Golden plays.** For Housing, tests added after the runs play five kinds of tenant on v14 worlds through the real market: a good lease whose accounting we checked by hand, a poor blind signing of a lemon, invalid and unauthorised actions (the market ends exactly as if the tenant had passed), malformed output through the kernel (typed as an error and scored as a pass, never as $0), and a world whose ceiling is $0. Each test has a counterexample that makes it fail.

F.3 Reading trajectories

Housing v14 (60 of its 1,800 attempts, step by step) and the agreement's development runs open in a trajectory viewer that shows each step as the seat saw it, beside the state it could not see, with a quality-control checklist for each run. On 2026-10-13 that checklist flagged 670 lines across 144 bundles, of which 58 needed a human reviewer. Tracing a single case by hand found Refund's defect in its payment leaf (Table F.4), and we traced an earlier data-center design the same way on 2026-09-14.

F.4 Adversarial checks applied

- **Trivial agents** are reported beside the models. Signing the draft unchanged recovers 14% of the value available in the agreement case's development set. Always paying the premium gives up $51.25 a world on Procurement's pre-registered worlds (Table 5.3). In Housing, scored after the runs, passing earns $0, signing without inspecting −$80 and an inspect-then-sign rule $145 a world, a rule that two of the three models do not beat (Table 5.2).
- **Fixed rules** are scored exactly against Procurement's best policy. In the earlier evaluation set (Table E.14) the one-step rule's loss of $0.06 is the finding that fails Gates 1 and 3. A v2 evaluation set built after the runs adds one mechanism: new listings of one seller kind share a source with a stated probability, so a cheap test of one listing informs the buyer about another. Included on what a buyer observes, with types drawn afterwards, its 24 worlds (from 5,798 seeds) make the one-step rule give up $30.09 a world on average and at least $10 in each world, as do the four fixed rules and a shortcut that ignores seller kinds. With the source never shared, the one-step rule comes within $0.33 of the best policy, so the gap is due to the shared source. Tests check every value against the environment's average over the 27 type profiles. Its separate 12-world gate check and pilot used 50 episodes ($0.2193). Gemini never tested a newcomer and matched the premium rule; Luna contacted a newcomer in 1 of 24 pilot plays, and neither model reached the planned probe. The required sizes were 326, 3,464 and 2,047 worlds, beyond the cap of 96, so no confirmatory run was frozen or played. On the harder evaluation set of Section 5.1, the same exact scoring places the one-step rule at the value of paying the premium in all 20 pre-registered worlds, and the shortcut and the rule that may test two newcomers at 33% and 82% of the gap (Table 5.3).
- **Audits of the task prompts** found instructions that decided outcomes in Procurement and a schema that contradicts its verifier in Refund.
- **Lower controls for Refund V2.5** were scored under a beatability rule committed before any control ran. For each seat and cohort, the reference must beat each control by at least 0.50 in end-to-end score (0.25 on

policy meaning in the policy seat), and a control may match the reference on at most 10% of cases. Nine controls played every public case, producing 3,468 trajectories with no model: empty, null, full-refund and random legal policies; an over-disclosing customer and a first-record customer; a first-sorted intake seat; and a payments seat that moves no money. The reference fails the rule on all 11 public cohorts in the policy, customer and payments seats. A full refund that collects nothing is fully right on 38 of 180 policy-seat cases, the evidence leaf scores any legal path as complete, the customer seat reduces to copying, and the payments seat reduces to how often the answer is no payment. Only the intake seat passes.

- **A placebo outcome** was run after the agreement's look-back price signal. The same regression on the opening price of a scripted integrator that is never told the hidden type gave 0.38 [0.23, 0.52] where the true value must be 0, which exposed hidden types tied to the evaluation set's design cell; a random relabelling of types had not caught this.

F.5 Statistical practice applied

The world is the unit of uncertainty, and comparisons are paired across conditions within worlds. On the agreement runs, treating plays as independent made the intervals 1.47 to 1.65 times too narrow, and adding worlds is more efficient than adding repeats: detecting a $50k gap needs 83 to 91 negotiations per arm at one play a world, against 159 to 162 at three plays a world. We name the counterparts and report results per counterpart type; in the agreement case the counterpart accounts for 25% to 33% of the spread in what a model earns. The reported analyses that were pre-registered before their runs are Housing v14 and v15, the agreement signalling test, both agreement confirmatory runs, and Procurement's test and its diagnostics. We tested a harness setting rather than assuming its effect: asking the CLIs for reasoning summaries changed nothing detectable in Housing (Opus +17 [−28, +62], Sol +3.5 [−46, +53]; five selected worlds and 120 episodes, with the summary-off plays always run first).

Review of the agreement's v1 confirmatory run (16 October). Gate 5 is `partial`. The run applied a 12-world variance pilot, a sealed sizing record, frozen plans, paired world-level intervals and Holm correction. Eight departures from the plan are documented: the CLI versions; an added wall-time stop; the run-plan labels; a change to the evaluation-set module for pilot generation; the permitted copy of the sizing code; a correction of an analysis label after the data were in; one missing negotiation in a secondary arm; and pilot pairing by play index. The label correction changes no estimate but leaves the analysis digest different from the sizing record, and the pinned verifier still needs manual treatment of the excluded receipt. These publication and configuration checks are therefore not fully closed. The client seat's construct limit is a separate issue. The 45-world estimate for the second primary contrast (P2) rejects zero (Holm p 0.0017), while its 98.3% interval [6.7, 46.1] lies below the pre-registered smallest meaningful difference of $50k; both readings must be reported.

F.6 Failure records and review

We record discovered failures in the incident log in four categories (design, operational, tooling and judgment), and we never delete rows (Table F.3). Evidence review by verification (digests, replay and a scan for prohibited text) was done for the Housing v14 bundle. The agreement family's registry change has had no review by anyone other than its author, and eight earlier changes to the shared runner were accepted without such a review under a one-time exception.

Table F.3. Incident-log entries about each case and its predecessors, by category (as of 16 October 2026).

Case	Design	Operational	Tooling	Judgment	Total
Housing	9	21	9	12	51
Procurement	27	6	19	5	57
Data center	37	22	22	9	90

Case	Design	Operational	Tooling	Judgment	Total
Refund	12	0	8	3	23

Counts are of distinct entries. One Data-center entry had been filed under an identifier already in use; we count it separately.

F.7 What quality control found

Table F.4. Twenty-five selected defects, the level at which each was visible, and what changed (dispositions as of 16 October 2026).

Case	What was wrong	Visible in one world?	How it was found	What changed
Housing	A seat-number tie-break and scripted rivals' $1 overbid decided who signed a sound listing	no: a pattern across seats	comparing signing rates by seat	one deciding tenant against pre-registered outside demand
Housing	The scripted landlord's accept-or-counter reply was a free, near-perfect quality test beside a $25 inspection	no: needs the cost distribution	validity audit before a second model ran	pooled landlord; the revealing landlord kept as a pre-registered arm
Housing	Inclusion made a favourite's lemon rate depend on its popularity (0.29 among included worlds where six tenants shared it, about half before inclusion)	no: a property of the included set	comparing rates before and after inclusion	include on the tenant's structure, read quality afterwards
Housing	The rule the tenant was told sat outside the sealed request	yes, by reading the sealed bytes	known at build, logged before any run	the rules sealed in the prompt, the tenant's history in the observation
Housing	A gap decomposition's "luck" part was not luck: uncontested blind bets were lemons 11 of 15 times, contested 1 of 15	no: a split by rival behaviour	testing the stated odds by model ($p = 0.006$)	blind signings scored at their expected value given what the tenant saw
Procurement (an earlier design)	A rule reading only the displayed price beat the qualified model by $28.50 a world	no: a baseline comparison	the construct check	construct gate recorded `failed`; Gate 0 made a profile mandatory
Procurement	The prompt pointed every subject at the cheapest listing, right in half the worlds and wrong in the other half	no: needs the evaluation set's construction	asking whether the hard half was valid or a trick	a neutral prompt and a new evaluation set
Procurement	The task prompt did not say a quote is final or that the buyer is judged on its average	no: a pattern in loss by action	listing each loss by the action that caused it	a second task prompt; loss fell from $12.2 to $1.0 and $19.4 to $5.4 a world
Procurement	Under sample noise the pre-award check reveals the true yield for one free action	yes, by reading the tool's code	designing a sample-again decision	open; 1 of 260 noisy episodes used it, no published number moves
Data center (predecessor case)	Walking away came within $11k to $46k of the right contract, and asking for every protection within $26k to $88k	no: a strategy-class simulation	simulating shortcut policies on the evaluation set	before any freeze: each named shortcut must cost $75k more than the best contract, and a deal must beat the better outside option by $150k
Data center (predecessor case)	At low effort the case could not separate two models: about 83% of the per-world gap was seed noise	no: needs paired seeds	variance split of the paired gap	one seed per world and more worlds; a redesign of the case
Data center	The scripted style hashed the whole world, including the client's hidden charge	no: needs two copies of a world	a test that the opening is identical for both types failed	the pre-registered test uses a fixed style; the included evaluation set draws the style from the public world only

Case	What was wrong	Visible in one world?	How it was found	What changed
Data center	The hidden type was fixed by design cell and moved with public quantities	no	a placebo outcome, after the look-back (Appendix F.4)	an included evaluation set whose types are drawn after inclusion (Appendix F.2)
Data center	The generator priced the client's rival offer, a public number in both task prompts, from the drawn integrator type, so a public parameter depended on both hidden types	yes, by reading the generator	building the included evaluation set	the included evaluation set prices it before any type is drawn; the development sets keep theirs
Data center	The integrator's private cost barely moves the best agreement, so the client seat has almost nothing to infer about clauses	no: a value of information across worlds	building the inclusion criterion for both seats	open; the client seat needs worlds where the integrator's risk charge is private
Housing	The primary measure conditions on the landlord's reply only below the sound floor; a reply that proves a listing sound is still scored at the stated odds	yes, by reading the measure's code	writing the formal appendix of this paper	recomputed from the sealed run roots: 57 signings move, no pre-registered verdict changes, Gemini's leads shrink by $10 and $13 in the v14 episodes; an erratum is owed
Refund	The 120-episode panel was six cases: the world seed changed only the case id, and the policy seat ran at temperature 0	no: needs the seeds compared	rebuilding the cases and reading the reference run's resolutions	the case is the unit of analysis; V2.5 varies pre-registered conditions by seed
Refund	A correct denial written with method `null` failed: the prompt says "none", the schema allows null, the verifier compares strings	yes, by reading one proposal against the verifier	building the gap decomposition	recorded; reported as format, not outcome
Refund	The human-QC approval was minted by the runner from two status-only stubs, so it bound whatever the runner produced	yes, by reading the runner's code	the Refund audit	V2.5 names a reviewer, but the runner still binds the approval to its own output (still open)
Refund	Refund's payment leaf checks that payments executed what the proposal asked, so a proposal with an aliased mutation moves no money where the reference refunds $185.22 and still scores 1.0	yes, by tracing one case	writing the formal appendix of this paper	open; report the leaf as binding and add a check against the reference
Housing	The v14 receipts' own scores total all six tenant seats against baselines from a market without outside demand, so they are not the deciding tenant's	yes, by reading the receipts	building the seat-0 controls	open; the published primary is computed separately and is unaffected; a seat-0 scorer needs a module outside the hashed runner
Refund	In three of Refund V2.5's four seats a naive control comes close to the reference: a full refund that collects nothing is fully right on 38 of 180 policy cases, the evidence leaf passes any legal path, the customer seat reduces to copying and the payments seat to the base rate of no payment	no: a rate across cases	lower controls under a rule committed before they ran	open; Gate 3 recorded `failed`; scoring pairs of cases would stop the full-refund control (0 of 38 pairs), a rule not yet pre-registered
Housing	The v14 receipts name the model for the five tenant seats a script played; the true seat map is only in an unsealed log	yes, by reading the driver's seat router	the paper's runtime review	open; record the seat map inside the seal under a new frozen configuration

Case	What was wrong	Visible in one world?	How it was found	What changed
Procurement	Procurement's original decision-loss report read an unsealed action copy through an unpinned reference; receipt replay alone did not certify it	yes, by reading the driver	the paper's runtime review	resolved for the reported panels on 16 October: source-pinned reanalysis from replay-audited sealed actions reproduces all 144 losses; prospective receipt-leaf integration is separate work
Procurement	The pay-or-test action schema let `message` be null while the parser requires it, so a model's first action could end its episode on format	yes, in one episode of the gate check	the flash-model gate	the schema requires every field the parser needs, under new frozen configurations

Fourteen of the 25 defects required evidence beyond a single world. Now that Procurement's decision-loss report has been re-derived from sealed actions, eight remain open: the free yield check in noisy Procurement evaluation sets; the Housing reply measure, the Housing receipts' score totals and their seat map; Refund's payment leaf, its runner-minted approval and its lower controls; and the Data-center client seat.

F.8 Assessment against the Agentic Benchmark Checklist

We assess the four cases item by item against the checklist (Zhu et al. 2025). The outcome-validity item that applies is the one for a quality measure, since no case is scored by string matching, unit tests or a judge model.

Table F.5. The four cases against the checklist.

Item	Status	Evidence
T.1 Tool versions specified	partly	routes, prompts, schemas and implementation digests are sealed in each plan; the two CLI agents expose no seed or temperature
T.2 Required APIs accessible	partly	provider outages lost episodes (36 Housing episodes to an empty account, 40 data-center negotiations to CLI errors), each recorded as typed missingness
T.3 Terminates or handles API errors	partly	Housing v14 uses worker-level failure streaks: twelve workers could still produce 36 account failures. Procurement flash panels declare retries and a consecutive-failure stop; the confirmatory agreement driver freezes a five-consecutive-failure halt plus list-price and wall-time ceilings, unlike its development predecessor. These are driver controls, not adoption of the kernel halt helper (Appendix H.4); operational failures remain exclusions
T.4 Residual state cleared between runs	met	each episode starts fresh from a sealed plan; the CLI agents run one ephemeral subprocess per action
T.5 Agent isolated from ground truth	partly	named hidden fields are tested absent from observations in Housing, Procurement and Refund, not in the agreement case; one leak was fixed and one is open in noisy-sample evaluation sets
T.6 Setup does not change over time	partly	environment inputs and pre-registered routes are pinned; this does not attest a stable hosted deployment, provider compliance, or the unsealed CLI environment (Appendix H.2, H.3)
T.7 Ground truth verified	partly	each reference is played on every included world through the environment and reaches its stated result, initially by tests added after development; the included agreement evaluation set also had a dry run before the confirmatory run (Appendix F.2)
T.8 Each task verified solvable	partly	reference policies are played through the environment, but full-information references can use information unavailable to the evaluated seat; this does not establish that every seat can attain the reported bound
T.9 Oracle solver	partly	exact for Procurement; an independent reference resolution and a scripted policy that resolves every case for Refund; a reachable reference for Housing, with an exact best informed tenant in development; a full-information stand-in for Data center, whose per-turn best informed policy is not built

Item	Status	Evidence
T.10 Free of exploitable shortcuts	partly	trivial agents, exact fixed-rule scores and task-prompt audits (Appendix F.4); twenty-five selected defects (Table F.4); a one-step rule reaches the best policy on Procurement's earlier evaluation set, and the reported evaluation set includes only worlds where that rule pays the premium; a wrong refund can still pass Refund's payment leaf
O.i.1 Quality metric resists exploitation	partly	code computes the measures and controls are tested, but the Housing measure needed correction and Refund has open payment and lower-control failures
R.1 Open source	partly	the environments, runner and evidence are public; the trajectory viewer is private
R.2 Open evaluation harness	met	the shared runner is in the public repository
R.3 Measures against contamination	met	preventive measures, not proof of zero contamination: worlds are generated, holdouts come from a disjoint seed range, and pre-registered tests run on fresh worlds
R.4 Plans to update	partly	evaluation sets regenerate from new seeds; no refresh schedule is stated
R.5 Capability related to construct	met	each case states the decision it measures (Table 5.1)
R.6 Evaluation subject stated	met	the subject is a profile (model, harness, prompt, tools, reasoning, retries); CLI agents are named with their harness
R.7 Steps to prevent, find and correct flaws	partly	the gates and the incident log (Appendix F.1, F.6)
R.8 Qualitative discussion of unavoidable flaws	met	the end of §5.2
R.9 Quantitative analysis of flaw impact	partly	measured for the task-prompt change, the refused episodes and missingness bounds; not for every open row
R.10 Statistical significance	met	paired world-level intervals; Holm correction across pre-registered primary contrasts
R.11 Guidance on interpreting flawed results	met	every result is labelled confirmatory, pre-registered or development
R.12 Non-AI baselines	partly	scripted policies in every case; no human
R.13 Trivial agents	partly	always paying (Procurement), signing the draft (Data center) and passing, signing anything and inspect-then-sign (Housing, scored after the runs) are reported; Refund V2.5 has nine lower controls, which come too close to the reference in three of four seats, and the V2.1 results reported here have none

Appendix G. Artifacts

Everything on which the numbers in Sections 5 and 6 rest is in the public AERead repository (https://github.-com/aeread-org/AERead), or is named here when it is not.

- **Code and cases.** The shared runner and the case families, with each case's generator, reference, verifiers, evaluation sets and tests. Each case is developed on its own branch of the repository.
- **Evidence.** Sealed, digest-bound evidence bundles: the Housing v14 confirmatory run and its pre-registered fill, the data-center agreement bundles, and the Refund V2.1 bundles with their gap decomposition, all on case branches. Procurement's pre-registered runs (Table 5.3) are on the main branch as twelve bundles, and all 432 of their receipts replay through the runtime's command line.
- **Quality control.** Each family's quality-control profile and incident log, kept with its code, and the reference-run tests named in §6.
- **Sources for every number.** The source files of Sections 5 and 6 end with tables that map each table and claim to its bundle, run, code revision or log entry.

- **Paper analyses.** A directory of paper analyses holds the rescoring under the corrected Housing measure, the failure taxonomy of §5.2 and the control contrasts of Table 5.2, with a runner and the inputs each analysis needs. The Housing v14 and Procurement inputs are sealed run roots not yet published as bundles, so those numbers can be checked against the run roots but not yet from the published evidence alone.
- **Not in the repository.** The trajectory viewer is private; one-page result summaries restate numbers from the bundles above.

Headline verification. The draft's source package includes the Housing reply-conditioned analysis and the Procurement reanalysis. The Housing analysis re-extracts its committed trajectory table from the sealed run roots and regenerates both measures exactly. The Procurement reanalysis checks all 144 original receipts and then reconstructs decision loss from their sealed parsed actions; its public projection regenerates Table E.14 without private run roots. The package supplies source and case digests, extraction commands, tests and a patch against the analysis base. Full verification of either projection against its original execution logs still requires the retained run roots.

Agreement confirmatory run and Procurement v2. The source package also includes the data-center agreement's confirmatory plan, results, analysis, sizing record and source-bound review, together with the Procurement v2 pilot report and results note. The original five agreement bundles retain their sealed bytes. We supply the corrections to the Data-center quality-control profile and incident log as a documentation patch; no frozen driver or analysis code was edited.

Appendix H. Runtime contract and experimental coverage

This appendix describes the runtime as of 8 October 2026 and the execution paths behind the reported results; we identify the later confirmatory run of the agreement case and the headline verification separately. The reported panels used different revisions, and the baseline does not retrospectively upgrade earlier execution or evidence. The full record definitions, adapter inventory and review history are retained in a technical supplement in the replication package. Appendix G identifies the experimental artifacts.

H.1 Execution contract

Families supply economic rules, observations, actions, transitions and scorers. The kernel resolves a plan, schedules play, records calls and actions, and finalizes evidence (Figure H.1). A resolved plan binds worlds and their content hashes, seat-to-profile assignments, sampling and pairing, prompts, agent settings, budgets, retry policies and implementation pins before provider execution. Hashes detect changes to the bound bytes; they do not establish that a provider obeyed a control or that a driver followed the specified seat assignment.

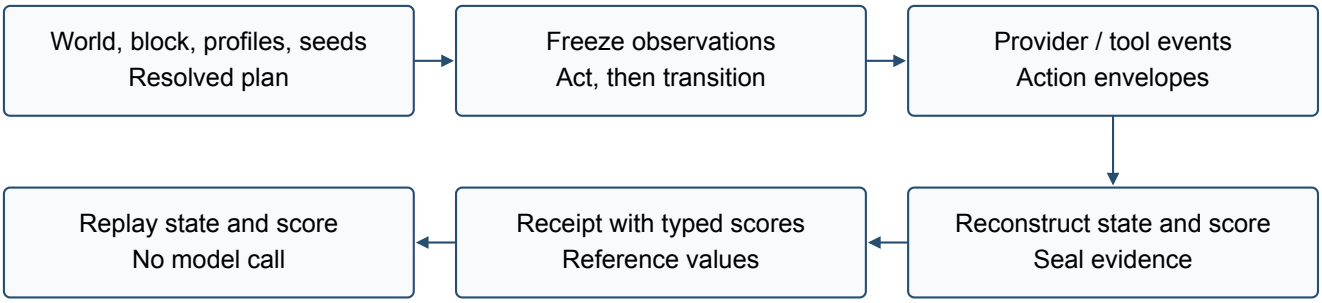


Figure H.1. The receipt path. Replay checks recorded transitions and receipt scores. It does not rerun the provider, harness or parser, or automatically cover analyses outside a receipt. The kernel supplies no sandbox or arbitrary-crash recovery.

For simultaneous play, the kernel freezes all observations before it requests any action, and the family receives the joint action set. This prevents a seat from observing a peer's same-phase action through request order; order effects inside the family transition remain the family's responsibility. Parsing and legality failures retain typed reasons. The invalid-action policy specified in the plan either stops execution or passes an invalid envelope to the family, which applies its stated consequence. An exception raised by a family hook is a defect in the evaluation apparatus, not a score of the agent under evaluation.

Each attempt has an append-only event log and content-addressed artifacts. Instrumented calls record a start before dispatch and a terminal outcome; actions, transitions and termination are also recorded. Finalization reconciles these records, reconstructs scoring input, invokes the scorer and seals the evidence before writing a receipt. Retries follow the conditions and limits specified in the plan. The October execution path rejects empty replies and incomplete truncated answers before family parsing; older executors can differ. These guarantees cover only the interfaces that the driver actually uses.

H.2 Scoring, replay and access

A receipt binds the plan, world, planned episode, attempt, per-seat profile hashes, implementation pins and evidence seal to typed score records. Measurement validity and inclusion in analysis are separate statuses. Each score records its metric, reference type, scope, direction and native units. Every applicable metric specified in the plan must be returned; deferred and inapplicable metrics are listed separately, and a deferred metric cannot serve as the primary score. At most one receipt is included per planned episode, but this cardinality check cannot establish that attempt selection preceded observation of the outcome.

Replay uses the family's state-transition and scoring hooks with recorded actions and compares reconstructed states and receipt scores. It reads the stored observation, response, parse result and legality result; it does not rerun the model, harness or parser. Completed scored executions, including those excluded as invalid measurements, carry full state-and-score replay. Operational-failure receipts carry no replay and are verified as exclusions. Replay neither converts an exclusion into a successful episode nor certifies an analysis outside the receipt.

At the October baseline, the runtime provides a reviewer command for replay (its usage is in the replication note). It checks published bindings against retained run evidence and reports verified, differing or missing-evidence rows. Coverage extends to every published episode only when the bundle identifies that trajectory grain; otherwise it covers the receipts found under the run root. Having this command available does not establish that it was run on a given panel.

Public live bundles contain sanitized receipt projections and digests; under the baseline policy they omit full prompts, raw responses, model reasoning, complete live receipts and failure messages. Independent replay therefore also requires retained run roots. A digest cannot recover omitted inputs. Appendix G records access

limits, and Table H.2 separates inline checks, later audits and headline-score reconstruction. Seeds and sealed requests support reconstruction of recorded play, without guaranteeing a fresh model response (He and Thinking Machines Lab 2025).

H.3 Applied controls and experimental coverage

The API and command-line paths do not apply the same controls. Table H.1 summarizes the command-line agents in the reviewed case studies. The OpenRouter client checks the reported model and route against the pinned route, but cannot independently attest the hosted deployment or whether a requested seed or reasoning control was honored. Reasoning-effort labels are not a common scale of deliberation across models.

Table H.1. Controls applied to the command-line agents in the reviewed panels. They describe the evaluated configurations, not all capabilities of the products.

Control	Claude Code / Opus 5.5	Codex CLI / Sol 6.1
Client and execution	The kernel's Claude Code client; one ephemeral process per action; low effort.	An external Codex client; one ephemeral process per action; low effort.
Sampling and output	Seed, temperature and output-token limits not applied.	Seed, temperature and output-token limits not applied.
Tools and session	Tools disabled; no persistent session.	Tool definitions retained; a call using a tool is refused.
Observed identity	Version 2.1.292; binary digest rechecked before calls.	Version 0.160.0; binary digest recorded; no served-model identity.
Accounting limits	Two internal model turns observed per kernel call; internal retries are not separate kernel attempts.	Adapter digest pinned; the tool's internal execution is not independently certified.
User customizations	`--safe-mode`: no CLAUDE.md, memory, skill, plugin, hook, output style or MCP server reached the model. The run's working directory, a repository checkout, and the account's email address were in context.	`--ignore-user-config` did not remove the operator's skills: eight were listed beside Codex's four built-in ones (about 500 input tokens a call), two described in benchmark and evaluation terms. The working directory's name contained the project's name.
Shared limits	Working directory and environment unsealed; costs estimated at list prices.	Working directory and environment unsealed; costs estimated at list prices.

Table H.1 describes the first confirmatory run of the agreement case. The second run isolated both tools and probed their context before and during its episodes, and every probe was clean (Section 5).

Table H.2 identifies the execution paths behind the reported results; the executor digests are in the replication note.

Table H.2. What produced the case-study numbers. Inline replay, a later audit and headline-measure reconstruction are separate checks.

Panel	Executor, driver and client	Receipt audit and headline coverage
Housing v14 1,800 planned episodes; separate 48-episode fill	October executor; run 6 October. Family price driver; OpenRouter on pinned routes, through the seat router; minimal-chat harness.	Original panel: 1,764 completed receipts and 36 operational-failure receipts. Inline replay, then bundle replay of 600 rows per model (564 scored for Gemini), and 48 fill rows. The receipt scores six tenants. The separately versioned seat-0 reanalysis was re-extracted from sealed records and regenerated exactly on 16 October.
Housing CLI 20 worlds, 40 episodes per agent	October executor; a Housing command-line driver. Claude Code and the external Codex client use the same seat router.	40 completed episodes per agent; inline receipt replay only. No published bundle or independent bundle-audit report in the review. Inherits Housing's seat and headline-score gaps. The driver is not digested in the plan.

Panel	Executor, driver and client	Receipt audit and headline coverage
Procurement CLI 24 worlds, two task prompts, two agents	October executor; a Procurement command-line driver; runs on 9 October. Same CLI clients as Housing.	24 receipts per run root. On 16 October, all 96 command-line receipts across both task prompts passed state and score audit; a source-pinned derivation from sealed actions reproduced every decision loss. Historical receipts still score contribution margin. Full run roots are retained.
Procurement flash 24 worlds per model	October kernel; the OpenRouter driver, digest recorded. Runs 13 October; the driver bytes match its committed version. Pinned OpenRouter clients.	24 completed receipts per model; 72 and 76 provider calls. On 16 October, all 48 receipts passed state and score audit and the pinned sealed-action derivation reproduced every decision loss. Table E.14's means and intervals are unchanged.
Data-center agreement development and confirmatory runs	Development: September executor; the agreement command-line driver. Confirmatory v1: its own driver, frozen plans, CLI versions and limits. Confirmatory v2 (Table 5.4): a driver built on v1's components, with isolation probes.	Development: 2,417 valid of 2,456, inline replay. Confirmatory v1 plus pilot: 1,175 valid of 1,176; another 120 scripted dry runs. All 1,295 scored receipts replay offline; one excluded receipt has a separate digest/seal check. Confirmatory v2: 640 valid of 640, every receipt replayed offline from its sealed events. Earnings and IR violations are receipt leaves.
Refund V2.1 four models, 120 episodes each in the reported panel	September kernel; the V2 runner; Arena client. Contributed registration calls the shared executor and finalizer.	Published projections carry state-and-score replay; inline replay. Gap figures are regenerated by the gap tool's check mode. Produces kernel receipts despite being outside the trusted scoring-contract fixture.
Refund V2.5 816 provider-free trajectories	A Refund release on the October runtime, without kernel changes. Contributed QC path; all seats use a scripted provider; paid calls refused.	Release replays and audits all 816 receipts. No published bundle in the review. These runs qualify release conformance, not paid-model performance; they do not replace the V2.1 results.

Housing: seat attribution and headline measure.

The v14 plan names the evaluated agent's profile for all six tenant seats, while a seat router in the driver supplies scripted policies for seats 1–5. The realized routing map is retained in an unsealed side log. The original receipt scores the sum over the six tenants, whereas the headline concerns seat 0, and the original headline reads outcome facts copied into a separate unsealed result file. The later reanalysis derives its corrected measure from sealed events. On 16 October, we re-extracted its trajectory table from the original sealed run roots and regenerated the committed analysis exactly. This remains a post-run, separately versioned derivation; the original receipt's certificate is unchanged.

Procurement: the decision-loss reference.

The receipt scores contribution margin and regret against a full-information bound. Headline decision loss instead uses the optimal policy under the buyer's information. The original report replayed an action trace copied into an unsealed result row through an unpinned reference implementation. On 16 October, a separate reanalysis audited all 144 original receipts through state and score replay, extracted parsed actions and validity from sealed events, and recomputed every loss with a source-pinned reference. The 96 final-brief results and 48 earlier-brief results match; preserving the original bootstrap order also reproduces Table E.14's intervals. The accompanying projection permits headline recomputation without private run roots, while re-auditing the original logs still requires them. This resolves the reported panels' derivation gap. It does not add a decision-loss leaf to their historical receipts or implement the prospective case-interface standardization that has been proposed.

H.4 Scope and remaining limits

The kernel supplies neither a sandbox nor complete mediation of external activity: families and tools can run outside its ports. Resume checks and continues a valid log, but a hard kill can leave an unmatched call start that cannot be finalized. An outcome-unknown record does not prove whether an external side effect occurred. Hosted-model determinism and arbitrary-crash recovery are not claimed.

Campaign stopping and spend accounting also depend on the driver. At the reviewed baseline no family driver used the kernel's optional halt on consecutive failures, and the kernel had no campaign-wide cost limit. The driver of the later confirmatory run of the agreement case froze its own five-failure halt, list-price cap and wall-time ceiling, and none of them triggered. Housing used worker-level stop rules; twelve workers account for its 36 refused episodes. The event-log spend reader distinguishes exact from incomplete totals, but had not been adopted by family drivers; CLI list-price estimates remain separate from billed charges. The inclusion rule does not prove pre-outcome attempt selection.

Integration and scientific qualification are distinct. The 21-package inventory includes wrappers, re-implementations and native packages with different fidelity, fixture and live-run coverage. The supplement retains those distinctions and the incomplete implementation-review status; package counts and passing tests do not establish case validity. Section 6 and Appendix F give the applied QC evidence and incident history. None of the four designs passes all of Gates 1–5 in this review, and later checks or repairs do not retroactively qualify earlier runs.